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PLATING POSTERS INC · METAL FINISHING REFERENCE SERIES

BORIC-SULFURIC ANODIZING

BSAA

PROCESS VARIANT **BSAA**

= REPLACES TYPE I CHROMIC

NO HEX CHROME

MIL TYPE IC · NO CR VI · AEROSPACE

BORIC-SULFURIC · THIN · AEROSPACE PAINT-BASE · NO HEXAVALENT CHROMIUM

THE COMPLETE TRAINING MANUAL

A full training course for the brand-new operator — from "wait, the part is the positive electrode?" to a signed certificate of completion. Thin, paint-ready boric-sulfuric anodize on aluminum — the aerospace finish developed to ***replace chromic-acid anodizing and get hexavalent chromium off the line entirely.*** *The good news:* ***no Cr VI carcinogen*** *in this bath. But it's still strong acid, caustic, hydrogen, and high-current DC — so we teach the chemistry and the discipline. Assumes you know nothing. Teaches you everything.*

01 INSP/RACK ▸ 02 CLEAN ▸ 03 ETCH ▸ 04 RINSE ▸ 05 DESMUT ▸ 06 RINSE

07 ANODIZE ▸ 08 RINSE ▸ 09 DYE ▸ 10 SEAL ▸ 11 RINSE/DRY ▸ 12 INSPECT

OPERATOR TRAINING & CERTIFICATION

EDITION 1.0 · 2026 · FOR-DUMMIES STYLE

Training reference only. Always verify exact parameters against your chemistry supplier's Technical Data Sheets (TDS), customer/aerospace specifications (e.g., MIL-A-8625 / MIL-PRF-8625 Type IC, Boeing BAC5632, and applicable AMS/customer process specs), and current Safety Data Sheets (SDS), and comply with OSHA, EPA, REACH, and local regulations before production use. **BSAA contains no hexavalent chromium — but sulfuric acid and caustic etch still cause severe burns, boric acid carries a reproductive-toxicity classification under REACH/CLP, hydrogen gas is flammable, and some seals run hot or contain chromate/fluoride.**

FIND YOUR WAY

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PART ONE — FRONT MATTER & FUNDAMENTALS

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★ **REMEMBER**

This manual is a training and reference tool — **not** a substitute for your facility’s written procedures, your supplier’s TDS/SDS, your customer’s spec (MIL-PRF-8625 Type IC, Boeing BAC5632, AMS, ASTM), or your supervisor’s instructions. When this manual and your shop’s official procedure disagree, **follow your shop’s official procedure and ask your supervisor.**

PART ONE — FRONT MATTER & FUNDAMENTALS

START HERE

WELCOME ABOARD

IF YOU DON'T KNOW AN ANODE FROM AN APRON, YOU'RE IN THE RIGHT PLACE — HERE THE WHOLE POINT IS AEROSPACE ANODIZING *WITHOUT* A CARCINOGEN

Welcome. If you are holding this manual, you are about to learn how to run — or at least understand — a **boric-sulfuric acid anodizing line**, the process the trade just calls **BSAA**. Maybe you started this week; maybe you've racked parts for a year and finally want to know *why*. Either way, you're in the right place.

Here is the first big thing to wrap your head around, and it's going to feel backwards if you've ever seen electroplating: **in anodizing, we are not coating your part with a different metal. We are growing a tough, ceramic oxide layer *out of the aluminum itself*.** And to do that, your part is hooked up as the **anode** — the *positive* electrode — the exact opposite of plating. That's the single biggest *process* idea in this whole book, and we'll teach it from the ground up.

Here is the second big idea, and it's the reason this process exists at all: **BSAA was invented to do the aerospace job that chromic-acid anodizing used to do — but without the hexavalent chromium.** Chromic anodize (Type I) is a fine, proven finish, but its bath is **Cr VI, a confirmed human carcinogen.** BSAA grows a similar thin, paint-ready, fatigue-friendly coating using **boric acid plus a small amount of sulfuric acid instead** — no hex chrome in the tank. That's why Boeing developed it and why so much of the industry has moved to it.

> ★ **REMEMBER**
> Do not read "greener and safer" as "harmless." BSAA removes the worst hazard (Cr VI), but the line still has **strong acid, hot caustic etch, flammable hydrogen, high-current DC, and (depending on your seal) hot tanks or chromate/fluoride chemistry.** Boric acid itself is low in acute hazard but carries a **reproductive-toxicity classification** you'll learn to respect.

The third thing up front: **this manual assumes you know nothing about anodizing, chemistry, or electricity.** That's not an insult — it's a promise. Every term gets defined the first time we use it; every step explains not just *what* but *why* — in the friendly, patient spirit of the *For Dummies* books. Don't memorize it; *understand* it.

• HOW TO USE THIS MANUAL

The manual follows the **line itself**, station by station, in the order a part actually travels: inspect/rack, clean, *lightly* etch, rinse, desmut, rinse, **anodize** (boric-sulfuric, part = anode, ~15 V), rinse, (almost never dye), seal *or leave unsealed for paint*, rinse/dry, inspect. Reading it in order matters: **the order of an anodizing line is not random** — and neither is this book's.

> 💡 **TIP**
> Read the Fundamentals (Part One) through *before* any station chapter — **especially Chapter 3, on why BSAA exists and how it differs from chromic.** The station chapters assume you already grasp the big picture: the part is the anode, oxide grows out of it, the coating is *thin* on purpose, and we got here by replacing a carcinogen.

THE ICONS — YOUR FRIENDLY MARGIN HELPERS

> 💡 **TIP**
> A shortcut or trick of the trade — the kind of thing a 20-year veteran would lean over and tell you.

> ★ **REMEMBER**
> A core idea worth locking in — the load-bearing ideas. If you forget everything else on the page, don't forget these.

> ⚠ **HEADS-UP**
> A safety warning. Read every single one — these protect your skin, eyes, lungs, and long-term health. We use them liberally.

> ⚙ **TECHNICAL STUFF**
> Deeper chemistry for the curious. Optional — skip it and you can still run the line.

> ☀ **FUN FACT**
> A bit of trivia to make the science stick (and to give you something to say at lunch).

TWO MORE THINGS YOU'LL SEE AT EVERY STATION

Each station chapter ends with a **Teach-Back** checklist (often paired with a red "Top Mistakes" card) — the things you must *say out loud, unprompted,* before you're cleared — and a bordered **Sign-Off** box your *trainer initials* once you've shown the skill safely. The Teach-Back is your study guide; the Sign-Off is the gate.

WHAT YOU'LL BE ABLE TO DO

LEARNING OBJECTIVES

MASTER THIS AND EVERY STATION CHAPTER CLICKS INTO PLACE

By the time you finish this manual, you should be able to confidently:

1. **Explain what anodizing is** in plain language — and how it is fundamentally *different from electroplating*: the part is the **anode**, and we **grow an aluminum-oxide layer out of the aluminum itself** instead of depositing a different metal onto it.
2. **State why BSAA exists:** it is the **boric-sulfuric, no-hexavalent-chromium replacement for chromic-acid anodizing (Type I)** — built to keep the aerospace-critical properties (thin coating, excellent paint/primer adhesion, good corrosion protection, minimal effect on fatigue strength) while eliminating the **Cr VI carcinogen.**
3. **Describe what the BSAA coating does** — a **thin (~2–7 µm), corrosion-protective oxide** that **barely changes dimensions, does not significantly reduce fatigue strength, and primes beautifully for paint and adhesive bonding.**
4. **Explain why BSAA is an *aerospace* anodize** — fatigue-critical structure, skins, fittings, faying surfaces, paint/primer bases — and recognize its place under **MIL-PRF-8625 Type IC**, **Boeing BAC5632**, and customer specs.
5. **Explain the thin porous oxide** — why anodic oxide has pores, why BSAA is usually **left as a paint base** rather than dyed, and the **seal choices** (dilute chromate vs. trivalent/non-chromate vs. unsealed for maximum paint adhesion).
6. **Explain dimensional growth** — the coating grows roughly half in/half out, but because BSAA is *thin*, the change is tiny — a key reason it's specified on tight-tolerance, fatigue-critical parts.
7. **Place BSAA in the family** — Type I (chromic, Cr VI, phasing out), **Type IC (non-chromic substitute — BSAA)**, Type II (sulfuric), Type III (hardcoat) — and recognize the related pretreatments **TSA** and **PAA**.
8. **Explain why aluminum alloy matters** — including the copper-rich aerospace alloys (2024, 7075) and clad sheet that BSAA is often run on.
9. **Name the stations of the line in order** and explain *why* the order can't be rearranged.
10. **Recognize the main hazards** — sulfuric acid (lower than Type II but still corrosive), boric-acid handling (low acute hazard but a REACH reproductive-toxicant), hot caustic etch, flammable hydrogen, DC electrical/arcing, and (depending on seal) hot tanks or chromate/fluoride — and know the correct PPE, emergency response, and waste handling.

> ★ **REMEMBER**
> You are not expected to hit all of these on day one. Anodizing is a craft. But the *safety* objectives (#10) are not optional and not gradual — you must have those before you work the line at all.

• THE LINE AT A GLANCE

Don't memorize yet — feel the rhythm: **Inspect/Rack → Clean → Etch (light) → Rinse → Desmut → Rinse → ANODIZE (boric-sulfuric) → Rinse → (Dye, rarely) → Seal (or unsealed for paint) → Rinse/Dry → Inspect.**

> ★ **REMEMBER**
> Notice the pattern: **a rinse follows almost every working tank.** Rinses aren't breaks — they're checkpoints that keep one chemistry from poisoning the next. On a BSAA line a bad rinse before the anodize tank or before the seal can quietly wreck the whole job.

> ☀ **FUN FACT**
> The aluminum oxide we grow in this tank is chemically the same family as **sapphire and ruby** (crystalline aluminum oxide, Al_2O_3). You're not bolting on a coating — you're turning the skin of the part into a ceramic that's part of the metal. And here we do it with **boric acid and a splash of sulfuric** — chemistry mild enough to be kind to fatigue-critical aircraft parts.

CHAPTER 1

WHAT IS ANODIZING?

THE TRUE BEGINNER'S PRIMER — SO FAR BACK WE DON'T EVEN ASSUME YOU KNOW WHAT ANODIZING IS

THE ONE-SENTENCE VERSION

Anodizing is using electricity to grow a hard, protective oxide layer out of the surface of an aluminum part — the part itself becomes part of the coating. That's the whole idea. Now let's unpack it, because it's genuinely different from plating and that difference trips up almost everyone at first.

FIRST, WHAT PLATING DOES (SO YOU CAN SEE THE DIFFERENCE)

In electroplating — zinc, nickel, chrome — you take your part, hook it up as the **cathode** (the *negative* electrode), and you *deposit a different metal* onto it from a bath. The coating is a *separate metal sitting on top of* your part.

Anodizing flips almost all of that:

- Your part is the **anode** — the **positive** electrode. (That's literally where the word "anodize" comes from.)
- We don't add a different metal. We take the **aluminum that's already there** and convert its surface into **aluminum oxide** by feeding electricity through an acid bath.
- The coating isn't sitting *on* the part — it **is** the part's surface, chemically grown and locked in. It can't peel like paint or flake like a bad plate, because it's integral to the metal.

> ★ **REMEMBER — THE SINGLE BIGGEST IDEA IN THIS BOOK**
> In plating, the part is the **cathode** and you **add** a metal. In anodizing, the part is the **anode** and you **grow oxide out of the part itself.** If you remember nothing else from Chapter 1, remember this. Everything else follows from it.

HOW IT PHYSICALLY WORKS: THE KITCHEN-SINK PICTURE

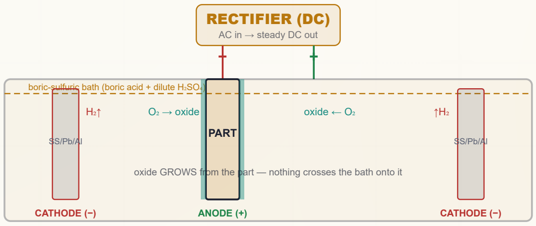
Picture a tank full of liquid called the **electrolyte** (or just "the bath"). For BSAA, that liquid is a **dilute mixture of boric acid and a small amount of sulfuric acid** in water. You hook two things to a power supply called a **rectifier** (it converts ordinary AC wall power into the steady one-direction "DC" current anodizing needs):

- **The anode** — this is your **aluminum part**, connected to the **positive** terminal. *"Anode = the part, the thing growing oxide."*
- **The cathode** — bars or plates (commonly **stainless steel, lead, or aluminum**) hanging in the same tank, connected to the **negative** terminal. The cathode just completes the circuit and is where **hydrogen gas** bubbles off.

> ⚙ **TECHNICAL STUFF — THE BORIC ACID'S JOB**
> Boric acid (H_3BO_3) is a weak acid that **buffers and moderates** the electrolyte. On its own, dilute sulfuric would keep dissolving and re-growing oxide aggressively (that's how Type II builds a thick, porous coating). Adding boric acid **tames** that chemistry so the coating grows **thin and dense at a low voltage (~15 V)** — close to what chromic acid produces, but with no chromium.

When you turn on the rectifier, here's the dance that happens:

1. Current flows. At the surface of your aluminum part (the anode), water in the bath is split and **oxygen** is delivered right at the metal surface.
2. That oxygen immediately combines with the surface aluminum to form **aluminum oxide (Al_2O_3)** — a hard, ceramic layer.
3. The acid is slightly aggressive, so as the oxide forms, the acid is *also* gently dissolving it — and this tug-of-war creates the famous **porous structure** (Chapter 4). In BSAA, the **boric acid moderates that tug-of-war**, so the coating stays thin and the bath is gentle on the metal.
4. Over at the cathode, **hydrogen gas** bubbles off into the air (a safety item — flammable gas — we'll come back to it).



Compare to a plating cell: the labels are reversed — here the **part is the ANODE (+)** and the counter-electrodes are the cathodes.

TECHNICAL STUFF — THE TWO REACTIONS

At the part (anode), aluminum is oxidized: simplified, $2Al + 3H_2O \rightarrow Al_2O_3 + 6H^+ + 6e^-$. At the cathode, those electrons reduce hydrogen ions: $6H^+ + 6e^- \rightarrow 3H_2\uparrow$. So **oxygen builds the coating at your part; hydrogen gas leaves at the counter-electrode.** No aluminum travels through the bath onto something else — it stays put and turns to oxide in place.

TECHNICAL STUFF — WHY BSAA RUNS WARMER THAN TYPE II

A standard Type II sulfuric tank must be held cold (~20 °C) to stop the concentrated acid from over-dissolving the oxide. BSAA uses **much less sulfuric acid** and adds boric buffering, so it is run **warmer — around 26–30 °C (≈80–86 °F)** — without burning the coating. Less refrigeration, lower voltage, shorter cycle: BSAA is also a *lower-energy* process than chromic. Cooling is still controlled (the reaction makes heat), just not the deep chill Type II/III demand.

FUN FACT

Anodizing is one of the few finishing processes where the coating is *grown from* the part rather than *added to* it. That's why anodized aluminum can't "chip off" to bare metal the way paint or a bad plate can — there's no glue line. The oxide and the metal are continuous.

CHAPTER 2

WHAT IS BSAA?

THE NO-HEX-CHROME AEROSPACE ANODIZE — A THIN, PAINT-READY, FATIGUE-FRIENDLY COATING

BSAA — boric-sulfuric acid anodizing — is a **thin aluminum-oxide coating** (typically **~2–7 µm / 0.00008–0.0003″**) grown in a bath of **boric acid plus a small amount of sulfuric acid**, at a **low voltage (~15 V)** and a **moderate temperature (~26–30 °C)**, in a **short cycle (~20–25 min)**. It was developed by **Boeing** (process spec **BAC5632**) as a direct, drop-in **replacement for chromic-acid anodizing (Type I)** — to keep the properties aerospace needs while **eliminating hexavalent chromium** from the tank. In **MIL-PRF-8625**, BSAA is the workhorse example of **Type IC: "non-chromic acid anodizing, to replace Types I and IB."**

• WHY AEROSPACE CHOSE IT

ADVANTAGE	WHAT IT MEANS ON THE FLOOR
No hexavalent chromium	The bath is boric + sulfuric — it removes the Cr VI carcinogen that chromic anodizing carries. Big win for worker health and the environment.
Thin coating, tiny dimensional change	Won't choke tight tolerances on machined fittings and fasteners.
Minimal effect on fatigue strength	The mild, buffered electrolyte and thin coating are kind to fatigue-critical structure (wing skins, spars, fittings).
Excellent paint / primer base	The porous oxide (especially unsealed) is one of the best primers for aerospace paint and adhesive bonding.
Good corrosion protection	Protects bare aluminum; even better when sealed with chromate/trivalent and/or painted.
Drop-in for chromic callouts	Many drawings that once called chromic now accept Type IC / BSAA.
Lower energy than chromic	Lower voltage, shorter cycle, less refrigeration.

WHERE YOU'LL SEE IT

Aircraft structural parts and skins, fittings and brackets, fasteners, faying (mating) surfaces in assemblies, and — above all — **anything that will be primed and painted.** BSAA is a *pretreatment and paint base*, not a decorative finish.

★ **REMEMBER**

BSAA's whole identity is **"the green replacement for chromic."** It does the *same aerospace job* — thin, fatigue-friendly, paint-ready, corrosion-protective — **without hexavalent chromium.** That one idea explains every parameter on this line.

⚠ **HEADS-UP — WHAT BSAA IS NOT**

BSAA does **not** give the thick, hard, wear-resistant coating of **Type II** (decorative/protective sulfuric) or **Type III** (hardcoat). It is **thin on purpose.** If a print wants color, abrasion resistance, or a heavy coating, BSAA is the wrong process — check the spec and ask.

💡 **TIP**

When a print says "Type IC," "BSAA," "boric-sulfuric," or cites **Boeing BAC5632**, that's *this* process. If it says "Type I" or "chromic," that's the older Cr VI process BSAA replaces — and BSAA may be an *approved substitute*, but only if your customer's spec allows it. Never substitute silently; confirm the callout.

☀ **FUN FACT**

BSAA came out of the aerospace industry's drive in the 1990s to get hexavalent chromium off the shop floor. It's one of a small family of "chrome-free" pretreatments — alongside thin-film sulfuric (TSA) and phosphoric-acid anodize (PAA) — that let modern aircraft be built with far less Cr VI than a generation ago.

CHAPTER 3 · READ TWICE

THE GREEN REPLACEMENT

WHY BSAA EXISTS — THE GOOD-NEWS SAFETY STORY, TOLD HONESTLY

This is the chapter that explains *why this line is here at all.* Slow down — it's the heart of BSAA's identity.

THE PROBLEM BSAA SOLVES

For decades, the standard aerospace anodize was **chromic-acid anodizing (Type I)**. It made a thin, paint-ready, fatigue-friendly coating — exactly what aircraft need. But its bath is **chromic acid, which is hexavalent chromium (Cr VI), a confirmed human carcinogen.** Cr VI is regulated under **OSHA 29 CFR 1910.1026**, its worst hazard is an *invisible mist* off the tank, and its waste must be chemically **reduced and precipitated** before disposal. Running a chromic line safely is possible but demanding — respirators, scrubbers, mist suppression, medical surveillance, and special waste treatment.

WHAT BSAA CHANGES

BSAA grows a **very similar coating** — thin, paint-ready, fatigue-friendly — but in a **boric-sulfuric bath with no chromium in it.** That single change cascades into a much friendlier line:

	CHROMIC (TYPE I)	BSAA (TYPE IC)
Bath chemistry	**Chromic acid (Cr VI)**	**Boric acid + dilute sulfuric (no Cr VI)**
Headline hazard	**Carcinogenic Cr VI mist**	Sulfuric acid + boric acid (no carcinogen)
Worker controls	Respirator, scrubber, mist suppression, Cr VI program	Standard acid PPE; no Cr VI program for the *anodize* tank
Voltage	Higher, ramped (up to ~40 V)	**Lower (~15 V)**
Temperature	Warm-ish	Moderate (~26–30 °C)
Waste	**Reduce-then-precipitate Cr VI**	Neutralize acid/boron; far simpler
Coating role	Thin, paint base, fatigue-friendly	**Same**

★ **REMEMBER — THE LOAD-BEARING IDEA**

BSAA was built to **keep the good (thin, paint-ready, fatigue-friendly coating)** and **remove the bad (hexavalent chromium).** That's the whole reason it exists. When someone asks "why not just run chromic?", the answer is: *because BSAA does the same job without a carcinogen in the tank.*

☀ **FUN FACT**

BSAA is sometimes called a "drop-in" because many shops converted their old chromic lines by **re-purposing the same tanks and rectifiers** — just swapping the chemistry and dialing the voltage down. Same footprint, far friendlier chemistry.

BUT “GREENER” IS NOT “HARMLESS” — THE HAZARDS THAT REMAIN

Here is the honest part. BSAA removes the *worst* hazard, but the line is still an industrial acid line. You must respect:

⚠ HEADS-UP — SULFURIC ACID (LOWER, BUT REAL)

The anodize bath still contains **sulfuric acid** — far less than a Type II tank (~30–50 g/L vs ~165–200 g/L), but still **corrosive.** It burns skin and eyes, and the tank can throw acid mist/spray. Full acid PPE; 15-minute flush for any contact; **add acid to water** for make-up.

⚠ HEADS-UP — BORIC ACID: LOW ACUTE HAZARD, BUT A REACH REPRODUCTIVE TOXICANT

Boric acid is **mild to the touch and not strongly corrosive** — but under **REACH/CLP it is classified as toxic for reproduction (Repr. 1B)** and is on the **Candidate List of Substances of Very High Concern (SVHC).** Translation: don’t create or breathe **boric-acid dust/mist**, don’t ingest it (no eating/drinking on the line, wash before breaks), and follow the SDS — especially for women of child-bearing age per your shop’s program. It is *not* a carcinogen and *not* in the league of Cr VI, but it is **not “harmless powder.”**

⚠ HEADS-UP — THE USUAL LINE HAZARDS STILL APPLY

Hot caustic etch (deep burns), **flammable hydrogen** at the cathode, **high-current DC** at contacts (LOTO), and — depending on your **seal** — a **near-boiling chromate or hot-water seal** (scald) or a **fluoride/nickel seal** (special first-aid). Each station chapter calls out its own.

⚙ TECHNICAL STUFF — WHY THE *SEAL* IS WHERE CR VI CAN SNEAK BACK IN

The classic aerospace seal for chromic *and* BSAA is a **dilute sodium-dichromate (Cr VI) seal**, because chromate sealing gives outstanding corrosion protection. If your shop uses a dichromate seal, **you still have Cr VI on the line — just in the seal tank, not the anodize tank.** Shops chasing *full* hex-chrome elimination switch to **trivalent-chromium (Cr III) or non-chromate seals**, or leave parts **unsealed** for painting. Know which seal your line runs (Chapter 11 / Station 10).

★ REMEMBER

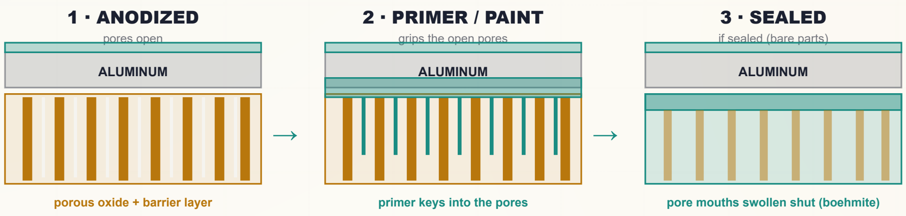
BSAA’s safety story in one line: **we took out the carcinogen, so respect the rest.** Acid, caustic, boron, heat, hydrogen, and electricity are all still here — and a chromate seal puts Cr VI back in one tank.

CHAPTER 4

THE THIN OXIDE

POROUS LIKE ALL ANODIZE — BUT HERE THE PORES ARE FOR *PAINT AND PRIMER*, NOT COLOR

Like every anodic oxide, the BSAA coating is **porous** — built of billions of tiny columns with a microscopic pore down the center of each, sitting on a thin, dense **barrier layer** against the metal. But BSAA uses those pores differently than a decorative line does.



WHY BSAA IS (ALMOST) NEVER DYED

Type II sulfuric anodize is famous for **dyeing** — color soaks into the open pores. BSAA *could* technically hold dye, but it almost never does, because the coating is **too thin** to hold deep, durable color; BSAA's job is a **paint/primer base and corrosion pretreatment**, not decoration; and aerospace parts get their color and protection from the **primer and topcoat applied over the anodize**, not from dye in the oxide. So on this line, **Station 09 (Dye) is usually skipped.**

★ **REMEMBER**

On a BSAA line the pores are for **adhesion, not color.** The open, slightly rough oxide is an excellent mechanical and chemical key for **paint, primer, and structural adhesive.** That's why BSAA parts are so often left **unsealed or only lightly sealed** before painting — an unsealed pore grips primer better than a sealed one.

THE SEAL TRADE-OFF, INTRODUCED

Sealing closes the pores. That **boosts corrosion resistance** of a *bare* (unpainted) part — but it can **reduce paint/primer adhesion.** So BSAA forces a choice the decorative lines never face: **going to be painted?** Often left **unsealed** (or lightly sealed) and primed promptly. **Staying bare** (or needing top corrosion protection)? **Sealed** — chromate, trivalent, or non-chromate. We cover this fully at Station 10. For now, lock in the idea: **on BSAA, "to seal or not to seal" is an engineering decision driven by whether the part gets painted.**

☀ **FUN FACT**

An unsealed, freshly anodized aluminum surface is so absorbent it will soak up a marker like blotting paper. On decorative lines that absorbency is for dye; on a BSAA line it's the same property that makes the surface drink up primer.

CHAPTER 5

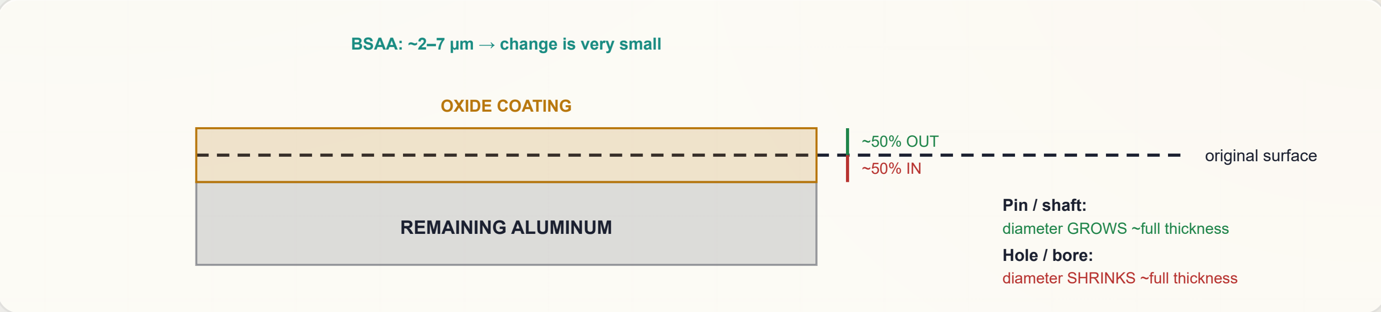
DIMENSIONAL GROWTH

THE COATING MAKES THE PART BIGGER — BUT HERE "ALMOST NONE" IS THE WHOLE POINT

Because we're converting aluminum into aluminum oxide, and oxide takes up **more room** than the metal it came from, the coating grows in **two directions at once**:

- About **half the coating thickness grows *into*** the original surface (consuming aluminum).
- About **half grows *out*** from the original surface (the part gets bigger).

For BSAA, the headline is that the coating is **thin (~2–7 µm)**, so the change is **tiny** — and that's *exactly why aerospace specifies it.* On a fatigue-critical, tight-tolerance fitting, a thin anodize that barely moves a dimension is a feature, not a limitation.



So for a coating that's, say, **5 µm (0.0002″) thick**, each surface moves outward by roughly **2.5 µm (0.0001″)**. On a pin or shaft, the **diameter grows by about the full coating thickness**; a hole's **diameter shrinks by about the full coating thickness** — but at BSAA thicknesses these are very small numbers.

★ **REMEMBER — THE 50% RULE (SMALL HERE, BUT REAL)**

Anodize grows **roughly half in, half out.** Each *surface* moves out about **half** the coating thickness; a *diameter* changes by about the **full** thickness. BSAA's thinness keeps the numbers small — but on precision aerospace fits you still plan for it.

💡 **TIP**

Because BSAA is thin and fatigue-friendly, it's often chosen *precisely* for parts where Type II would remove or add too much. Still, confirm the print: critical fits, bearing surfaces, and threads may be **masked** or machined to allow for growth.

⚠ **HEADS-UP — MASKING AND ELECTRICAL CONTACT**

Anodize is **electrically insulating.** Any surface that must conduct after finishing (a ground path, a bonding/grounding pad) must be **masked** so no oxide forms there. Threads, bearing fits, and faying surfaces that need bare metal are masked or controlled per the drawing.

⚙ **TECHNICAL STUFF — THE PILLING–BEDWORTH IDEA**

Aluminum oxide occupies more volume than the aluminum it formed from (Pilling–Bedworth ratio > 1). That expansion is *why* the coating grows outward — and the same expansion lets a hot-water seal swell the pores shut. One fact (oxide is bulkier than metal) explains both dimensional growth *and* sealing.

CHAPTER 6

THE ANODIZE FAMILY

SAME IDEA, SEVERAL FLAVORS — KNOW EXACTLY WHICH ONE YOUR PRINT IS ASKING FOR

All these processes grow an integral aluminum-oxide coating with the part as the anode. They differ in **acid, thickness, and purpose.**

	TYPE I — CHROMIC	TYPE IC — BSAA *(THIS MANUAL)*	TYPE II — SULFURIC	TYPE III — HARDCOAT
Electrolyte	Chromic acid (**Cr VI**)	**Boric + dilute sulfuric**	Sulfuric (~165–200 g/L)	Sulfuric (concentrated/cold)
Typical thickness	~0.5–7.5 µm	**~2–7 µm**	~5–25 µm	~25–100+ µm
Voltage	Higher, ramped	**~15 V (~10–22 V)**	~12–18 V	Higher V
Temperature	Warm-ish	**~26–30 °C**	~20–21 °C	Cold (~0–10 °C)
Cr VI in bath?	**Yes**	**No**	No	No
Main use	Aerospace, fatigue, paint base	**Aerospace paint base / pretreatment — chromic replacement**	Decorative + protective	Maximum hardness/wear

★ **REMEMBER**

Type I = chromic (Cr VI, phasing out). BSAA = Type IC, boric-sulfuric, the no-Cr-VI replacement for Type I. Type II = sulfuric, decorative + protective. Type III = sulfuric hardcoat, thick, maximum wear. This manual is **BSAA / Type IC** — the green aerospace anodize.

💡 **TIP — THE TWO COUSINS YOU'LL HEAR ABOUT**

Beyond the numbered types, two related **chrome-free aerospace pretreatments** sit near BSAA:

TSA (Thin-Film Sulfuric Anodize): a *dilute* sulfuric anodize, also developed as a chromic replacement — thin, for corrosion protection and paint base. A close sibling of BSAA; some customers spec one or the other.

PAA (Phosphoric-Acid Anodize): a phosphoric-acid anodize optimized for **structural adhesive bonding** (aircraft bonded assemblies). It's the bonding specialist — **we build PAA next in this series.** Each gets its own manual; all stand on the same anodizing fundamentals you're learning now.

⚙ **TECHNICAL STUFF — WHY THREE "CHROME-FREE" PRETREATMENTS, NOT ONE**

Chromic did several jobs at once (corrosion pretreatment, paint base, bonding prep). Replacing it without Cr VI took a small *family* — **BSAA** and **TSA** for general anodize/paint-base duties, **PAA** for the most demanding **adhesive bonding.** Which one a part gets depends on the customer's qualified process spec, not on operator choice.

CHAPTER 7

ALUMINUM ALLOYS

ANODIZING ONLY WORKS ON ALUMINUM — AND AEROSPACE RUNS THE TRICKIEST ALLOYS

Anodizing only works on **aluminum** and a handful of other “valve metals” (titanium, magnesium, tantalum, niobium — each with its own process). On this line it’s aluminum. But “aluminum” isn’t one thing — it’s dozens of **alloys**, and the alloying elements change how the part anodizes.

THE AEROSPACE REALITY: COPPER-RICH ALLOYS

- **2xxx series (copper)** — e.g., **2024.** Strong, fatigue-tough, but **copper makes them harder to anodize cleanly** and more prone to defects. This is a big reason BSAA’s gentler, buffered chemistry matters — it’s kinder to copper-bearing alloys than a hot concentrated sulfuric bath.
- **7xxx series (zinc, plus copper)** — e.g., **7075.** Very high strength; anodizes reasonably but can vary; sensitive to over-etching.
- **Clad (“Alclad”) sheet** — 2xxx/7xxx core with a thin **pure-aluminum cladding** for corrosion resistance. The pure-Al surface anodizes beautifully — but the cladding is *thin*, so **over-etching can cut through it** and expose the copper-rich core. **Light etch discipline (Station 03) protects the cladding.**

THE FRIENDLY ALLOYS

5xxx (magnesium) and **6xxx (magnesium-silicon, e.g., 6061/6063)** anodize **very well. 1xxx (pure)** is excellent and clear. **Cast, high-silicon** alloys come out **gray/mottled.**

ALLOY FAMILY	MAIN ELEMENT	ANODIZE BEHAVIOR
1xxx (pure Al)	none (99%+ Al)	Excellent, very clear
2xxx	**Copper**	Harder to anodize well; common in aerospace — BSAA handles it more gently than hot sulfuric
5xxx	Magnesium	Very good
6xxx (6061/6063)	Mg + Si	Excellent
7xxx	**Zinc (+Cu)**	Good but variable; aerospace structural
Clad (Alclad)	pure-Al skin	Anodizes great — **don’t etch through the thin cladding**
Cast (high-Si)	Silicon	Gray/mottled

★ **REMEMBER**

Aerospace alloys are the hard ones. Copper (2xxx) and zinc-copper (7xxx) and **clad sheet** are exactly where BSAA earns its keep — its mild, buffered, low-voltage chemistry is **gentle on the base metal and on fatigue life.** Know your alloy *before* you process, and treat clad parts with light-etch care.

💡 **TIP**

Always confirm the **alloy and temper** on the traveler. The same recipe on 2024-T3 versus 7075-T6 versus clad sheet can behave differently. When in doubt, process a sample of the *actual* alloy/lot first.

☀ **FUN FACT**

The aluminum cladding on Alclad sheet is a corrosion *sacrifice* layer — pure aluminum protecting a copper-rich core. It also happens to anodize cleaner than the core, which is why etch control on clad aerospace skins is a genuine craft.

CHAPTER 8

HOW THE LINE WORKS

WALKING THE LINE — EVERY TANK HAS A JOB, AND THE ORDER IS THE RECIPE

A BSAA line is a sequence of tanks. Here's the journey and *why each step has to be where it is.*

1. **Inspect / Rack** — Check the part, confirm alloy and spec, and clamp it tight to a rack. The part **is the anode**, so a loose contact = no coating or a burned spot. The contact point won't anodize, so it's chosen carefully.
2. **Clean / Degrease** — Remove oils and shop soil. Oxide won't grow evenly through grease.
3. **Etch (light)** — A *controlled, light* caustic dip removes the natural oxide skin and a *little* aluminum, evening the surface. **Light**, because these are fatigue-critical, often clad parts — we remove as little metal as possible.
4. **Rinse** — Wash off caustic before it contaminates the acid steps.
5. **Desmut / Deox** — Etching leaves a dark **smut** (alloying elements — especially **copper** on aerospace alloys — that didn't dissolve). An acid deox removes it for a bright, uniform surface.
6. **Rinse** — Clean before the main tank.
7. **ANODIZE** — The main event. Part = anode, **boric-sulfuric bath**, **ramp to ~15 V**, ~26–30 °C, ~20–25 min. Grow the thin oxide.
8. **Rinse** — Gently rinse acid from the fresh, open pores.
9. **Dye (rarely)** — Usually skipped on BSAA (paint base, not decorative).
10. **Seal — *or not*** — Close the pores (chromate / trivalent / non-chromate) **for corrosion**, *or* leave **unsealed** for maximum **paint/primer adhesion.** An engineering decision (Station 10).
11. **Rinse / Dry** — Final clean-up and gentle dry.
12. **Inspect** — Thickness, seal/adhesion as specified, appearance; and **prompt move to paint** if unsealed.

★ **REMEMBER — WHY THE ORDER CAN'T MOVE**

You can't desmut before you etch (no smut yet). You can't anodize over grease or smut (defective oxide). You can't seal before you'd dye (if dyeing). And if the part is going to paint, you handle the **seal decision** to protect adhesion. The line *is* the recipe.

⚠ **HEADS-UP — THE LINE IS A TOUR OF HAZARDS**

As you walk the line you pass **hot caustic** (etch), **acids** (desmut, boric-sulfuric anodize), **boric acid** (handle dust/mist with care), **flammable hydrogen** (cathode), **high-current DC** (rectifier/contacts), and — depending on seal — a **hot or chromate/fluoride seal.** Each station chapter calls out its own.

💡 **TIP**

When you're learning, walk the empty line with your trainer and just name each tank's *job* and *main hazard* out loud. If you can do that for all twelve, the rest of the manual is mostly detail.

CHAPTER 9

PPE — YOUR DAILY ARMOR

WHAT YOU WEAR, AND WHY EACH PIECE IS THERE

A BSAA line works with sulfuric acid, boric acid, hot caustic etch, high-current DC, hydrogen gas, and (often) a hot or chromate seal. The good news: **no Cr VI in the anodize tank** means no respirator/scrubber regime *for the anodize bath itself*. The rest of the line still demands proper gear.

HAZARD ON THE LINE	PPE THAT PROTECTS YOU
Sulfuric acid (anodize, desmut) — burns, splash	Sealed chemical **splash goggles** + **face shield**, chemical-resistant **gloves**, **apron/suit**, chemical-resistant **boots**
Boric acid — low acute, but a reproductive toxicant	Gloves; avoid **dust/mist**; no eating/drinking; wash before breaks; per SDS and shop program
Hot caustic etch — deep burns	Same as acid PPE; caustic burns are deep and slow to feel
Hot seal (chromate / hot water ~90–100 °C) — scald/steam	Face shield, heat-resistant gloves, long sleeves; mind the steam
Chromate seal (if used) — Cr VI in *that* tank	Per your shop's Cr VI program for the seal tank: gloves, goggles, ventilation, hygiene
Fluoride seal/desmut (if used) — deep, delayed burns	Gloves/goggles; know **calcium gluconate** first-aid per SDS
DC electrical / arcing at contacts	Dry hands, dry gloves, no jewelry, follow LOTO
Slips on wet floors	Non-slip chemical-resistant boots

💡 **TIP**

Put your gear on *in order*: boots and apron first, then gloves, then goggles, then face shield last. Take it off in reverse. This keeps you from touching your face with contaminated gloves. Rinse your gloves *before* you take them off.

★ **REMEMBER**

On a BSAA line your **eyes and skin** are the most exposed. Sulfuric *and* hot caustic *and* (often) a hot seal all live here. **Sealed splash goggles and a face shield are not optional** at the working tanks — even though the carcinogen is gone.

⚠ **HEADS-UP — GOGGLES, NOT JUST SAFETY GLASSES**

Safety glasses stop flying chips; they do **not** stop an acid or caustic splash from the side. At every chemical tank, sealed splash goggles (plus a face shield when charging, dumping, or handling parts) are required.

⚠ **HEADS-UP — NEVER EAT, DRINK, SMOKE, VAPE, OR APPLY COSMETICS ON THE LINE**

Hand-to-mouth contact is a major route for ingesting chemicals — **and boric acid's reproductive-toxicity classification makes ingestion control especially important.** Wash thoroughly before breaks and before leaving. Keep food and drink completely out of the work area.

CHAPTER 10

EMERGENCY & FIRST-RESPONSE

KNOW THESE COLD BEFORE YOU TOUCH A TANK

Know these *before* you need them. In an emergency, seconds matter and you won't have time to read.

SITUATION	WHAT TO DO — IMMEDIATELY
Acid or caustic on skin	Flush immediately with water for **at least 15 minutes.** Remove contaminated clothing while flushing. Get medical attention — caustic burns especially can be deep even when they don't hurt at first.
Acid or caustic in eyes	Go straight to the **eyewash**, hold eyes open, flush **at least 15 minutes**, get medical help. Don't stop early.
Large splash / drench	Use the **safety shower** — full 15 minutes, clothing off.
Hot seal-tank scald	Cool the burn with water; do not pop blisters; get medical attention for anything beyond minor.
Boric acid ingestion / large dust exposure	Get away from the source, rinse mouth, **do not eat/drink**; follow SDS first-aid and get medical advice — note its reproductive-toxicity classification.
Acid spill	Don't play hero — alert others, contain only if trained, use the spill kit, follow your shop's spill plan. Never mix acids and caustics during cleanup.
Electrical / arc at rectifier or contacts	Don't touch — de-energize via rectifier controls / LOTO. Never reach into a tank with the current on.
Hydrogen gas	Keep ignition sources (flames, sparks, grinding) away from the cathode/tank area; ensure ventilation is running.
Fluoride exposure (fluoride seal/desmut, if used)	Fluoride burns are deep, delayed, and systemically toxic — flush, then apply the **specific first-aid** in the SDS (often **calcium gluconate gel**) and get medical help immediately. Ordinary flushing alone may not be enough.
Chromate-seal splash (if used)	Treat as Cr VI: flush thoroughly, follow your shop's Cr VI exposure procedure, get medical attention.

⚠ **HEADS-UP — THE GOLDEN RULE FOR ACID MAKE-UP**

Always add ACID to WATER, never water to acid. ("Do as you *oughta*: add acid to water.") Adding water to concentrated acid can flash-boil and erupt it out of the container. Bath make-up is usually authorized-personnel work — but every operator must know this rule.

⚠ **HEADS-UP — LOCKOUT/TAGOUT (LOTO)**

LOTO is mandatory for *all* rectifier and electrical maintenance. Physically lock the power off and tag it so no one can switch it back on while someone is working. "I'll just be a second" is how people get electrocuted.

★ **REMEMBER**

Know where the **nearest eyewash, safety shower, and spill kit** are at *every* station before you start your shift. In an acid/caustic emergency you won't have time to look. **Eyewash and shower stations must always be unblocked.**

CHAPTER 11

THE SAFETY PHILOSOPHY OF A BSAA LINE

THE MINDSET THAT TIES EVERY STATION TOGETHER

1. WE REMOVED THE WORST HAZARD — NOW RESPECT WHAT'S LEFT.

The biggest safety win of BSAA is **no hexavalent chromium** in the anodize tank. Celebrate that — then stay sharp, because acid, caustic, heat, hydrogen, and electricity are all still here.

2. TWO CORROSIVES, MIRROR-IMAGE.

The **etch is caustic** (high pH); the **desmut and anodize are acid** (low pH). Both burn. Treat high pH and low pH with equal respect — and never let them mix outside a controlled neutralization.

3. BORIC ACID IS MILD TO THE TOUCH BUT NOT TO IGNORE.

Low acute hazard, but a **reproductive toxicant** under REACH/CLP. Control **dust, mist, and ingestion.** Hygiene matters.

4. THE PART IS LIVE.

Because the part is the anode, there is **DC current at the work** and at the contacts. Wet hands + current + acid is a bad combination.

5. HOT IS A HAZARD TOO.

A chromate or hot-water **seal runs near boiling.** Steam scalds. And if your seal is chromate, that tank carries Cr VI even though the anodize tank doesn't.

6. HYDROGEN IS FLAMMABLE; ENGINEERING CONTROLS FIRST.

The cathode bubbles off hydrogen gas — ventilation on, ignition sources away. Tank exhaust, temperature control, lids, and splash guards do the heavy lifting; PPE is your last layer, not your first. And when in doubt, **stop and ask** — the unsafe operator is the one who guesses.

• THE HAZARD FAMILIES TO INTERNALIZE

1. **Corrosives (acid AND caustic).** Sulfuric (anodize/desmut) and caustic (etch). *Defense:* goggles, face shield, gloves, apron; 15-minute flush; acid-to-water.
2. **Boron + (possible) chromate/fluoride in the seal.** Boric-acid hygiene; if chromate seal, Cr VI controls in *that* tank; if fluoride, calcium-gluconate first-aid. *Defense:* SDS literacy, gloves, hygiene, ventilation.
3. **Hot / thermal.** Near-boiling seal, hot etch, hot cleaner. *Defense:* face shield, heat gloves, steam awareness.
4. **Electrical & gas.** High-current DC; flammable hydrogen. *Defense:* LOTO, dry hands, no ignition sources, ventilation.

★ **REMEMBER**

BSAA's safety story is **"we took out the carcinogen, so respect the rest."** If you can name the hazard family at the tank in front of you, you already know most of what you need to protect yourself.

★ **REMEMBER**

You now have the whole foundation: what anodizing is (and how it's *not* plating), the part-is-the-anode/grow-oxide idea, *why BSAA replaced chromic*, the thin paint-base oxide, dimensional growth, the family, alloys, how the line works, what to wear, and the safety mindset. Every station chapter builds on this.

PART TWO — STATION BY STATION

STATION 01 OF 12

INSPECTION & RACKING

“IN ANODIZING, THE PART IS THE WIRE. A LOOSE PART IS A DEAD PART.”

PURPOSE & THE “WHY”

Inspect the incoming aluminum part, confirm the **alloy, temper, and spec** (BSAA / Type IC, thickness, sealed/unsealed, paint-to-follow), and **rack it** so it makes firm, current-carrying contact and hangs correctly. Because the part **is the anode**, every bit of current that grows your coating flows *through the rack contact into the part.* A loose or corroded contact means high resistance — **no coating, a thin coating, or a burned contact spot** — and possibly a dropped part in an acid tank. Racking also decides *where* the un-anodized contact mark lands, how gas escapes, and how solution drains.

★ **REMEMBER — THE CONTACT MARK IS UNAVOIDABLE**

Wherever the rack grips the part, **no oxide grows** (that spot must carry current). So you place the contact where the mark won’t matter — a hidden face, an edge, a hole — and you make it **tight.** On aerospace parts, the contact location is often specified; follow the drawing.

ITEM	SPEC / PRACTICE	WHAT TO WATCH
Incoming inspection	Per traveler / print	Confirm **alloy & temper**, Type IC/BSAA, thickness, seal state, paint-to-follow
Rack material	Titanium or aluminum	Ti racks don’t anodize away; Al racks anodize and need stripping
Contact pressure	Firm, spring-loaded / bolted	Loose = burning, no coat, dropped parts
Contact location	Hidden/non-critical area (often spec’d)	The contact point won’t anodize (bare mark)
Masking	Plugs/tape/lacquer on no-anodize areas	Threads, grounds, bearing fits, faying/bonding surfaces
Clad parts	Handle gently	Thin cladding — avoid gouges that expose the core
Orientation	Allow gas escape & drainage	Avoid trapped air pockets (un-anodized voids)

⚠ **HEADS-UP**

Racks and parts have **sharp edges and pinch points**, and you’ll handle them near chemical tanks. A part that drops because of a weak rack becomes a **splash hazard** in an acid or caustic tank, not just a scrap part.

💡 **TIP — TITANIUM RACKS ARE WORTH IT**

Titanium racks passivate rather than anodize, so they hold good contact load after load and don’t need stripping. Aluminum racks anodize along with the part — insulating the contact over time — so they must be **stripped/re-pointed** regularly.

STEP-BY-STEP

1. Read the traveler (alloy, temper, BSAA/Type IC, thickness, masked areas, seal/paint plan). 2. Inspect for scratches/gouges/prior finish — anodize *reveals* flaws. On clad parts, watch for cladding damage. 3. Plan a hidden contact point (per drawing) that can carry full current. 4. Mask no-anodize areas per print. 5. Rack firmly — good metal-to-metal contact, correct orientation. 6. Verify no parts touch each other (shadowing). 7. Handle by the rack from here on.

SIGN-OFF — STATION 01

Teach-Back: Why contact matters more in anodizing than plating (part = anode); why the contact point won’t anodize; why titanium racks are preferred; why orientation (gas/drainage) matters; two no-anodize areas that get masked and why; why clad parts need gentle handling.

“I verified the alloy, temper, and spec, chose and made a firm contact point in a non-critical (spec’d) area, masked all no-anodize features, racked for gas escape and drainage with no parts touching, and handled parts by the rack.”

Trainee: ____________________ Trainer: ____________________ Date: _____________

STATION 02 OF 12

CLEAN / DEGREASE

“ANODIZE OVER OIL AND YOU’VE GROWN A DEFECT, NOT A COATING.”

PURPOSE & THE “WHY”

Get the part **chemically clean** — no oil, grease, fingerprints, or shop soil — so the oxide grows evenly. Oxide grows uniformly only from clean aluminum; oil and soil cause **uneven anodizing and patchy adhesion.** Cleaning is usually a **mild alkaline (non-etching) soak cleaner** — formulated *not* to attack the aluminum (we save controlled metal removal for the etch). The free, foolproof check is the **water-break test.**

TIP — THE WATER-BREAK TEST (FREE, FAST, NEVER LIES)

Pull a cleaned part and watch the water. A truly clean surface sheets water in **one continuous, unbroken film** (PASS). A dirty surface makes water **bead up** (FAIL — re-clean). Costs nothing; takes two seconds; use it on every part.

```
FAIL - water beads up        PASS - water sheets evenly
+--------------+             +--------------+
|  o   o   o   |  X          |##############|  OK
|    o    o    |             |##############|
+--------------+             +--------------+
Oil remains -> RE-CLEAN      Surface clean -> proceed
```

PARAMETER	RANGE	WHAT TO WATCH
Type	Mild **non-etching alkaline** cleaner	Must not attack aluminum (or clad skin)
Temperature	~50–70 °C (120–160 °F)	Per cleaner TDS
Time	~3–10 min	Heavy oil longer
Concentration	Per TDS	Titrate regularly
pH	Mildly alkaline	Still a skin irritant/burn risk
Water-break test	Pass = continuous sheet	The only reliable field check

HEADS-UP — HOT ALKALINE SOLUTION

Even a “mild” cleaner is hot and alkaline: splashes irritate/burn skin and eyes, and steam irritates airways. Splash goggles, gloves, apron. Skin → flush 15 min; eyes → eyewash 15 min and get help.

STEP-BY-STEP

1. Check tank temperature and concentration in range. 2. Immerse the racked load fully; start the timer; confirm agitation. 3. Withdraw slowly, draining back into the tank. 4. Run the **water-break test** — continuous sheet → proceed; beading → re-clean. 5. Move promptly to the etch (don’t let a clean part sit).

SIGN-OFF — STATION 02

Teach-Back: What the cleaner removes (oil/soil) vs. what it must *not* do (attack aluminum — that’s the etch); correctly perform/read a water-break test; the caustic/irritant hazard and PPE.

“The part passed the water-break test, the cleaner was in range and at temperature, and I wore required PPE.”

Trainee: ____________________ Trainer: ____________________ Date: ______________

STATION 03 OF 12

ETCH (LIGHT)

"WE REMOVE AS LITTLE METAL AS POSSIBLE — THESE PARTS FLY, AND THE CLADDING IS THIN."

PURPOSE & THE "WHY"

Remove the natural oxide skin and a *minimal, controlled* layer of aluminum in a **caustic etch**, giving a clean, uniform surface for anodizing. Unlike the decorative Type II line — which etches heavily for a matte satin look — **BSAA etches *light*.** These are **fatigue-critical, tight-tolerance, often clad** aerospace parts. Every micron of aluminum you remove can cost dimension, and on **clad sheet** over-etching can cut through the thin pure-aluminum cladding and expose the copper-rich core. **Light etch protects fatigue life, tolerance, and cladding.**

★ **REMEMBER — LIGHT, NOT HEAVY**

On BSAA, the etch is a *cleaning and activation* step, not a cosmetic one. **Minimal metal removal** is the goal. Over-etching here is a real defect: lost tolerance, reduced fatigue life, and (on clad parts) breached cladding.

PARAMETER	RANGE	WHAT IT DOES / WATCH
Chemistry	**Sodium hydroxide**, often with additives	Dissolves oxide + a little aluminum
Concentration	Per spec (often milder than decorative)	Higher = faster/more aggressive
Temperature	~50–65 °C (120–150 °F)	Hot — burns and speeds etch
Time	**Short** (controlled, per spec)	Longer = more metal loss — avoid
Result	Clean, uniform surface + **smut**	Smut is normal — removed at desmut

⚠ **HEADS-UP — HOT CAUSTIC IS ONE OF THE WORST BURNS ON THE LINE**

Caustic burns are **deep, get worse with time, and may not hurt much at first.** Full PPE: sealed goggles, face shield, elbow-length chemical gloves, apron, boots. Skin → flush **15+ minutes**; eyes → eyewash 15 min and get medical help immediately.

⚠ **HEADS-UP — CAUSTIC + ALUMINUM MAKES HYDROGEN**

Caustic attacking aluminum releases **hydrogen gas** (flammable). Keep ignition sources away and ventilation running.

STEP-BY-STEP

1. Confirm concentration and temperature in range. 2. Immerse; start the timer for the **specified (short) etch time.** 3. Agitate gently / per procedure. 4. Withdraw and drain back into the tank — the part will look **dark/smutty** (normal, especially on copper alloys). 5. Move **directly to rinse** — don't let caustic dry. **Never extend etch time to "clean up" a part — re-route it instead.**

SIGN-OFF — STATION 03

Teach-Back: Why BSAA etches *light* (fatigue, tolerance, clad cladding); what the etch removes (oxide + minimal aluminum); why the part comes out dark/smutty and what fixes it (desmut); the caustic-burn hazard and first response.

"I etched for the specified short time at the correct concentration/temperature, minimized metal loss to protect fatigue/tolerance/cladding, moved straight to rinse, and wore full PPE for hot caustic."

Trainee: ______________ Trainer: ______________ Date: __________

STATION 04 OF 12

RINSE (THE FIREWALL)

"A RINSE ISN'T A REST STOP. IT'S A CHECKPOINT."

PURPOSE & THE "WHY"

Wash the **caustic film** off the part before it reaches the acidic desmut and anodize chemistry. The etch is caustic (high pH); desmut and anodize are acidic (low pH). Carry caustic drag-out forward and it **neutralizes and contaminates the acid baths**, wastes chemistry, and causes staining. The rinse knocks drag-out down by dilution. Cleanliness is tracked by **conductivity** (µS/cm) — *lower = cleaner.*

TIP — COUNTER-FLOW RINSING

The professional approach is a **counter-current** cascade: parts move forward, water flows backward, so the cleanest water touches the part last. It saves enormous water — important on a line with this many rinses.

PARAMETER	RANGE	NOTES
Temperature	Ambient	No heating needed
Time	30–60 s with agitation	Longer for cavities/blind holes
Water source	Municipal or DI	DI preferred near anodize/seal
Conductivity	< 500 µS/cm (target < 300)	Monitor at shift start
Flow	Continuous overflow / counter-current	Prevents stagnation

HEADS-UP

This rinse carries dragged-in caustic and starts mildly alkaline. Goggles, gloves, non-slip footwear. **Wet floors and slips are the #1 everyday injury at rinses.**

DO

- Check conductivity at shift start; tell your lead if near/above limit.
- Transfer promptly from the etch — don't let caustic dry.
- Immerse fully with agitation 30–60 s (longer for cavities).
- Lift and drain over the rinse tank before moving on.

TOP MISTAKES

- Treating the rinse as a quick dunk.
- Ignoring conductivity.
- Letting parts drip-dry between etch and rinse.
- Not flushing cavities; skipping the drain step.

SIGN-OFF — STATION 04

Teach-Back: Why carrying caustic into the acid stages causes problems; define drag-out / drag-in; what conductivity tells you and which way is "clean."

"Rinse conductivity was within limit, water flow was running, and parts were rinsed the full time with PPE worn."

Trainee: ____________ Trainer: ____________ Conductivity: ________ µS/cm

STATION 05 OF 12

DESMUT / DEOX

"ETCHING UNCOVERS THE ALLOY'S LEFTOVERS — AND AEROSPACE ALLOYS LEAVE A LOT OF COPPER."

PURPOSE & THE "WHY"

Remove the dark **smut** the etch left behind — the alloying elements (especially **copper**, plus silicon, iron) that didn't dissolve in caustic — using an **acid desmut/deox**, leaving a bright, uniform surface for anodizing. Anodize over smut and you get **dark streaks, dull patches, and poor coating/adhesion.** On **copper-rich aerospace alloys (2024, 7075)**, the etch leaves a heavy copper-bearing smut, so desmut here matters even more than on a decorative line. The desmut chemistry depends on the alloy: simple alloys may need only **nitric** or **sulfuric**; high-copper/silicon alloys often need a **fluoride-bearing or proprietary deox.**

★ **REMEMBER**

Smut after etch is normal — and heavier on copper alloys. Desmut's whole job is to remove it. A part that enters the anodize tank still gray/smutty will come out streaky and dull. Desmut is the bright-surface gatekeeper.

PARAMETER	RANGE / PRACTICE	NOTES
Chemistry	Nitric and/or sulfuric; fluoride-bearing/proprietary deox for high-Cu/Si	Match to the alloy
Concentration	`Per product TDS`	Titrate regularly
Temperature	`Often ambient-warm`	Per TDS
Time	`~0.5–3 min`	Until smut is gone, no longer
Result	Bright, uniform, active surface	No gray/copper film remaining

⚠ **HEADS-UP — STRONG ACID (AND POSSIBLY FLUORIDE)**

Desmut is acidic; **fluoride-bearing desmuts are especially hazardous** (hydrofluoric-type chemistry — deep, delayed burns, systemically toxic). Know exactly which chemistry your tank uses, read its SDS, wear full acid PPE, and know the **specific first-aid** (fluoride burns may require **calcium gluconate** per your SDS/medical plan). **Add acid to water** for any make-up.

STEP-BY-STEP

1. Confirm which desmut chemistry the tank is (and its specific hazards/first-aid). 2. Check concentration and temperature in range. 3. Immerse for the specified short time — just until smut clears. 4. Withdraw, drain, move to rinse. 5. Don't over-dip — excessive desmut can etch/pit some alloys and (on clad) attack the cladding.

SIGN-OFF — STATION 05

Teach-Back: What smut is and where it came from (alloying elements left by the etch — heavy copper on aerospace alloys); why anodizing over smut causes defects; that desmut chemistry depends on the alloy; the special hazard/first-aid of a fluoride-bearing desmut.

"I removed the smut with the correct desmut for the alloy, did not over-dip, and understand the specific hazard and first-aid for this tank (including fluoride if applicable)."

Trainee: ____________ Trainer: ____________ Chemistry: ________

STATION 06 OF 12

RINSE (PRE-ANODIZE)

“THE LAST CLEAN HANDOFF BEFORE THE MAIN EVENT.”

PURPOSE & THE “WHY”

Remove desmut acid before the part enters the anodize tank. Carrying desmut drag-out into the anodize tank introduces unwanted ions (and, with a fluoride desmut, **fluoride contamination** that can roughen/pit the coating). A clean rinse protects the boric-sulfuric bath chemistry — and a clean, consistent handoff keeps coatings uniform part-to-part.

PARAMETER	RANGE	NOTES
Temperature	Ambient	—
Time	30–60 s, agitation	Longer for cavities
Water	DI preferred	Cleaner handoff to anodize
Conductivity	Per shop spec	Lower = cleaner
Flow	Counter-current overflow	—

HEADS-UP

Same rinse rules: drag-in acid, wet floors, goggles/gloves. If the prior tank was a **fluoride** desmut, this rinse water carries fluoride — handle and dispose accordingly.

TIP

Move from this rinse into the anodize tank **promptly and consistently.** A long, variable dwell in air or rinse before anodizing can cause part-to-part thickness/coating differences. Consistency is quality — and it matters on flight hardware.

STEP-BY-STEP

- Confirm flow and conductivity.
- Rinse the full time with agitation.
- Drain over the tank.
- Move straight to anodize.

TOP MISTAKES

- Quick dunk instead of a real rinse.
- Inconsistent dwell before anodize.
- Ignoring conductivity.
- Forgetting fluoride drag-in handling.

SIGN-OFF — STATION 06

Teach-Back: Why a clean handoff to the anodize tank matters; what fluoride drag-in could do.

“The part was fully rinsed, conductivity was acceptable, and I moved promptly and consistently to the anodize tank.”

Trainee: ____________ Trainer: ____________ Date: ________

STATION 07 OF 12 · THE MAIN EVENT

ANODIZE — THE MAIN TANK

WHERE THE ALUMINUM'S OWN SURFACE BECOMES A THIN, PAINT-READY CERAMIC — WITH NO HEXAVALENT CHROMIUM

Take a breath. Up to now, every station has had one job: get the part clean, even, and ready. **Station 07 is where the anodize is actually created** — the tank that defines this whole line, and the tank that does the chromic job *without* chromium.

The big idea in one sentence: **we run the aluminum part as the *anode* in a boric-sulfuric bath and ramp the DC voltage to about 15 V to grow a thin, paint-ready aluminum-oxide layer out of the part's own surface.** The cast:

- **Anode** — *your aluminum part* (**positive**). Where the oxide grows.
- **Cathode** — **stainless steel, lead, or aluminum plates** (negative). They carry current and bubble off hydrogen; they do **not** feed metal into the part.
- **Electrolyte** — **boric acid (~5–10 g/L) + dilute sulfuric acid (~30–50 g/L)** in DI water, held at **~26–30 °C (≈80–86 °F)**, with agitation.

★ **REMEMBER — THE ANODIZE TANK IS THE MIRROR IMAGE OF A PLATING TANK**

In plating, the part is the **cathode** and metal lands on it. Here the part is the **anode** and **oxide grows out of it.** The cathodes don't supply the coating — all of it comes from the part's own aluminum plus oxygen from the bath. Forget this and the whole tank behaves like a mystery.

★ **REMEMBER — AND THIS TANK HAS NO CR VI**

Unlike the chromic Type I tank this process replaced, the BSAA bath contains **no hexavalent chromium.** That's the entire reason BSAA exists. There's no carcinogenic mist to scrub here — but it's still strong acid, DC, and hydrogen, so respect remains mandatory.

⚠ **HEADS-UP — SULFURIC ACID + BORIC ACID + DC + HYDROGEN, ALL AT ONCE**

This tank holds **sulfuric acid** (corrosive; less than Type II but real) and **boric acid** (a reproductive toxicant — control dust/mist), runs **high-current DC** (shock/arc at contacts), and bubbles **flammable hydrogen** off the cathodes. Engineering controls — **ventilation, temperature control, agitation, splash control** — must be working before you run it. Full acid PPE. No ignition sources.

• WHY BORIC-SULFURIC (AND WHAT MAKES IT PARTICULAR)

ADVANTAGE	WHAT IT MEANS ON THE FLOOR
No Cr VI	The headline — chromic's carcinogen is gone
Thin, paint-ready oxide	Excellent primer/adhesive base (Station 10 / Part Three)
Fatigue-friendly	Mild buffered chemistry, thin coat — kind to flight hardware
Low voltage / lower energy	~15 V, short cycle, modest heating
Drop-in for chromic	Many Type I callouts accept Type IC / BSAA

…but the same chemistry demands care: **voltage and ramp must be controlled** (the coating and its properties depend on it), the bath is **sensitive to chloride and fluoride contamination** (pitting), **dissolved aluminum and boric/sulfuric balance** must stay in range, and it grows the coating **half into the part** (small, but real, dimensional growth).

• THE BATH MAKE-UP — KNOW THE RANGES

⚠ **HEADS-UP**

These are **typical** BSAA ranges for training context. **The controlling document is your customer/process spec — Boeing BAC5632, MIL-PRF-8625 Type IC, and your shop's lab sheet.** Verify before you run; never anodize to a number from a training manual.

COMPONENT / PARAMETER	TYPICAL RANGE	WHAT IT DOES / WATCH
Sulfuric acid (H_2SO_4)	~30–50 g/L (~3–4.5% v/v)	The active electrolyte — **far less than Type II's 165–200 g/L**
Boric acid (H_3BO_3)	~5–10 g/L	**Buffers/moderates** the bath → thin, dense coating at low V
Temperature	~26–30 °C (≈80–86 °F)	Moderate — warmer than Type II; controlled, not deep-chilled
Voltage (DC)	**Ramp to ~15 V** (~10–22 V)	**Voltage-controlled** process; hold at the spec voltage
Ramp rate	Controlled (per spec)	Ramp up smoothly to target voltage
Coating thickness	~2–7 µm (0.00008–0.0003")	Thin by design (paint base)
Cycle time	~20–25 min (ramp + hold)	Per spec
Cathodes	Stainless / lead / aluminum	Carry current; bubble H_2; do not feed metal
Agitation	Air or solution agitation	Even temperature & fresh electrolyte at surface
Chloride contamination	Very low	Causes **pitting/burning** — use DI, keep it out
Fluoride contamination	Very low	Drag-in from fluoride desmut roughens/pits
Dissolved aluminum	Per spec	Too much dulls/streaks & shifts the bath

★ **REMEMBER — THE NUMBERS TO CARRY**

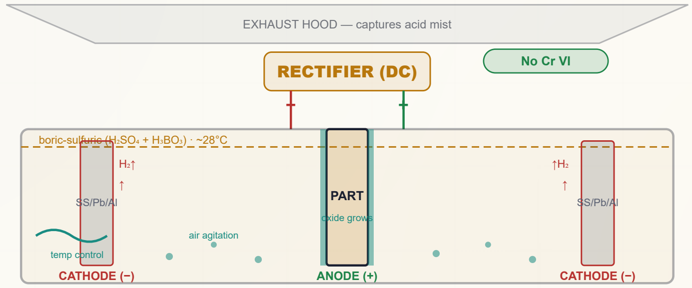
Boric ~5–10 g/L + sulfuric ~30–50 g/L, **~26–30 °C**, **ramp to ~15 V**, **~2–7 µm**, **~20–25 min.** Note how different this is from Type II: **much less sulfuric, lower voltage, warmer, thinner, shorter, voltage-controlled.**

⚙ **TECHNICAL STUFF — VOLTAGE-CONTROLLED, NOT CURRENT-CONTROLLED**

Type II is usually run at a set **current density** for a calculated time (the "720 rule"). **BSAA is typically run at a set *voltage* (~15 V).** You ramp up to the target voltage and hold it for the cycle; current naturally tapers as the insulating oxide forms. Thickness is governed by **voltage and time** within the spec'd window — don't expect to "push more amps for a thicker coat" the way you would on Type II.

⚙ **TECHNICAL STUFF — CONTAMINATION IS THE SILENT KILLER**

Chloride (from tap water, sweat, drag-in) and **fluoride** (from a fluoride desmut/seal dragged in) both cause **pitting and burning** even at low levels. Use **DI water**, rinse well, and keep contamination out. Watch **dissolved aluminum** too — as it builds, the bath dulls and shifts and eventually needs partial replacement.



Key params: `H2SO4 30-50 g/L · H3BO3 5-10 g/L · 26-30C · ramp to ~15V · 2-7um · agitation`. Note the **reversed polarity vs. a plating cell:** the part is the ANODE (+).

• STANDARD OPERATING PROCEDURE

1. **Confirm the part is clean, etched (light), desmutted, rinsed, and racked tight** — straight from the rinse, no long air dwell.
2. **Verify engineering controls FIRST:** temperature in range (~26–30 °C), agitation on, ventilation on, splash control in place, full acid PPE on.
3. **Check bath readiness:** sulfuric and boric in range (per lab), temperature in range, chloride/fluoride/dissolved-aluminum within limits, cathodes present/sound.
4. **Confirm rectifier polarity: PART = ANODE (POSITIVE).** *(Opposite of plating — verify every time.)*
5. **Set the voltage ramp** to reach the spec target (~15 V) at the spec ramp rate.
6. **Hold at voltage for the spec'd cycle time** (~20–25 min) for the target thickness.
7. **Monitor** voltage, current taper, temperature, and agitation through the run.
8. **At the end of the cycle, stop current, withdraw, drain over the tank,** and move to the post-anodize rinse — the pores are now open and ready for **prompt sealing or priming.**
9. **Log everything:** voltage, current, time, temperature, thickness checks, additions.

> ⚠ **HEADS-UP — VERIFY POLARITY EVERY SINGLE TIME**
> Because everyone trained on plating expects the part to be the **cathode**, the most dangerous habit on this line is assuming polarity. **The part is the ANODE (+).** Reversed polarity gives no coating (and can damage parts/cathodes). Check it before you energize.

COMMON DEFECTS — WHAT YOU'LL SEE AND WHAT IT MEANS

WHAT YOU SEE	MOST LIKELY CAUSE	WHAT TO DO
Thin / no coating	Bad contact; **wrong polarity**; low voltage; weak bath	Re-rack tight; **verify part = anode**; check voltage/bath
Powdery / soft coating	Bath too warm/out of balance; over-aggressive	Check temp & chemistry; verify agitation
Burned/rough at edges & contact	Local current crowding; poor contact; chloride	Improve racking; check chloride; check ramp
Pitting	**Chloride or fluoride contamination**; impurities	Lab-check Cl^-/F^-; use DI; treat bath per chemist
Dull, gray, streaky	Copper alloy (2024/7075); smut not removed; high dissolved Al	Confirm alloy; verify desmut; lab-check Al
Poor paint/primer adhesion later	Over-sealed for paint; contamination; aged surface; over-etched	Review seal decision; prime promptly; check cleanliness
Voltage won't hold / current odd	Poor contact; failing cathodes; bath imbalance	Check contacts/cathodes; lab-check bath

INTEGRATED SAFETY — RESPECT REMAINS (EVEN WITHOUT CR VI)

⚠ **HEADS-UP — SULFURIC + BORIC (THE HEADLINE PAIR)**

Sulfuric acid is corrosive (splash/mist) — lower concentration than Type II but real. Boric acid is a **reproductive toxicant** — control dust/mist/ingestion. Engineering controls first, PPE always; skin/eye contact → flush **15 minutes**; **acid to water** for make-up.

⚠ **HEADS-UP — HIGH-CURRENT DC & FLAMMABLE HYDROGEN**

Shock/arc at contacts and bus bars — **LOTO** for all electrical work, dry hands only. Hydrogen off the cathodes is **flammable** — ventilation on; no flames, sparks, or grinding nearby.

⚠ **HEADS-UP — THE GOOD NEWS, STATED PLAINLY**

There is **no hexavalent-chromium mist** at this tank. That removes the single worst hazard of the chromic process this replaced. **Do not let that breed complacency** about the acid, boron, electrical, and hydrogen hazards that remain.

TEACH-BACK — CAN YOU ANSWER?

- Which electrode is your part, and which charge? *(Anode, +.)*
- Where does the coating come from? *(The part's own Al + O_2.)*
- The headline numbers (boric, sulfuric, temp, voltage, thickness, time)?
- Why is this process voltage-controlled?
- What does the boric acid do? Why is there no Cr VI here?
- The remaining safety families?

TOP MISTAKES TO AVOID

- Assuming the part is the cathode (it's the **anode**).
- Assuming Type II numbers apply (BSAA is much milder).
- Running with bath out of balance/temperature.
- Loose rack contact; ignoring chloride/fluoride.
- Letting anodized parts age before seal/prime.
- Eating/drinking near the tanks (boric acid).

SIGN-OFF — STATION 07

"I can independently: verify temperature, agitation, ventilation, and PPE before running; verify bath readiness (boric, sulfuric, temp, chloride/fluoride/Al per lab); confirm the part is racked as the ANODE (positive); ramp and hold the correct voltage; control thickness by voltage and time; recognize thin-coat/pitting/burning defects; and state the acid, boric-acid, electrical, and hydrogen hazards — and that this tank has no Cr VI. I understand chemistry adjustments are made only by authorized personnel."

Operator: ____________________ Date: __________ Trainer: ____________________ Date: __________

STATION 08 OF 12

RINSE (POST-ANODIZE)

“GENTLE NOW — THE PORES ARE WIDE OPEN, AND PRIMER IS COMING.”

PURPOSE & THE “WHY”

Rinse the boric-sulfuric electrolyte out of the freshly anodized part **without contaminating or prematurely sealing the open pores**, before seal or paint. The fresh anodic oxide is **porous and absorbent** — it will soak up whatever it touches. Leftover acid interferes with sealing and adhesion. The rinse must be **clean (DI)** and, if the part will be sealed later, **not too hot** (hot water can start sealing the pores prematurely).

★ **REMEMBER — DON’T PRE-SEAL BY ACCIDENT**

After anodizing, **use cool/ambient DI rinse** before any sealing decision. Hot rinse water here can start closing the pores. Save the heat for the actual seal step (if one is used).

PARAMETER	RANGE	NOTES
Water	`DI strongly preferred`	Pores absorb impurities
Temperature	`Cool / ambient`	Avoid premature sealing
Time	`1–3 min, gentle agitation`	Flush acid from pores
Conductivity	`Low (per spec)`	Confirms acid removed

⚠ **HEADS-UP**

This rinse carries dragged-out **sulfuric/boric** electrolyte — mildly acidic, slip hazard, goggles/gloves. Don’t let acid/boron-laden rinse go untreated to drain (Part Three / waste).

STEP-BY-STEP

- Confirm DI flow, cool temperature.
- Rinse gently, full time.
- Drain.
- Move to seal (or, if unsealed for paint, handle cleanly toward priming).

TOP MISTAKES

- Using hot water before sealing (pre-seals pores).
- Using hard tap water (impurities in pores).
- Short rinse leaving acid behind.

SIGN-OFF — STATION 08

Teach-Back: Why the pores are absorbent now; why cool DI; what hot water would do prematurely; why prompt handling toward seal/prime matters.

“I rinsed with cool DI to flush electrolyte from the open pores without pre-sealing, and moved promptly to seal or toward priming.”

Trainee: ____________________ Trainer: ____________________ Date: _____________

STATION 09 OF 12 · USUALLY SKIPPED

DYE (USUALLY NOT USED)

“ON A PAINT-BASE ANODIZE, COLOR COMES FROM THE PRIMER — NOT FROM DYE.”

PURPOSE & THE “WHY”

On decorative lines (Type II), this is where color soaks into the open pores. **On a BSAA line, dyeing is almost always skipped.** BSAA is a **thin pretreatment and paint base**, not a decorative finish — its coating is too thin to hold deep, durable color, and the part’s color and protection come from the **primer and topcoat applied over the anodize.** Most BSAA parts go from the post-anodize rinse **straight to seal (or straight toward priming if unsealed).**

★ **REMEMBER**

BSAA’s pores are for adhesion, not color. If a print actually calls for a *dyed* anodize, that’s almost certainly a **Type II** job, not BSAA — verify the spec and ask. Don’t dye a BSAA part on assumption.

💡 **TIP — THE RARE EXCEPTION**

Occasionally a thin anodize is lightly dyed for **identification** (e.g., to visually distinguish processed parts or lots) where a customer spec allows it. Even then, follow the exact spec — many aerospace specs **prohibit** dye on a bonding/paint surface because it can interfere with adhesion. **No dye unless the spec explicitly calls for it.**

⚠ **HEADS-UP — IF YOU EVER DO DYE**

Some dyes are skin/eye irritants or sensitizers; dye baths may be warm and mildly acidic. Gloves and goggles, read the SDS. And remember: if dyed, the part still must be sealed **before** it can be considered finished — and sealing then competes with paint adhesion (Station 10).

⚙ **TECHNICAL STUFF — WHY THE THIN COAT CAN’T HOLD COLOR WELL**

Because BSAA runs at low voltage in a buffered bath, its coating is thin with a relatively substantial **barrier layer** and only a shallow porous layer. That structure is great for adhesion and corrosion-as-pretreatment, but it is *not* the deep, dye-friendly sponge a thick Type II coating is. Different tool, different job — which is exactly why dye lives on the Type II line, not here.

SIGN-OFF — STATION 09

Teach-Back: Why BSAA is normally *not* dyed (paint base, thin coat); that color/protection come from primer/topcoat; when (rarely) dye is allowed and why most paint/bond specs prohibit it.

“I confirmed the spec does not call for dye, skipped this station, and moved the part toward seal or priming.”

Trainee: ____________________ Trainer: ____________________ Date: ______________

STATION 10 OF 12

SEAL (OR UNSEALED FOR PAINT)

"THE BIG BSAA DECISION: CLOSE THE PORES FOR CORROSION — OR LEAVE THEM OPEN FOR PAINT."

PURPOSE & THE "WHY"

Decide and execute the **seal state** of the BSAA coating. Sealing **closes the open pores**, which **boosts corrosion resistance of a bare part and locks the coating down** — but it can **reduce paint/primer and adhesive bonding.** So BSAA forces a real engineering choice that the decorative lines never face:

- **Going to be painted/primed?** Often **left unsealed (or only lightly sealed)** and primed promptly — for **maximum adhesion.**
- **Staying bare, or needing top corrosion protection? Sealed** — and then you choose *which* seal.

★ **REMEMBER — THE BSAA SEAL RULE**

Seal = corrosion protection on a bare part, but weaker paint adhesion. Unsealed = best paint/bond adhesion, but less protection while bare. The right choice is set by the **drawing/spec and whether the part gets painted** — it is *not* an operator preference. Confirm the seal callout every job.

THE SEAL OPTIONS

SEAL OPTION	TEMPERATURE	CR VI?	TRADEOFFS
Dilute chromate (sodium dichromate)	`hot, ~90–100 °C`	**Yes (Cr VI)**	Excellent corrosion; the traditional aerospace seal — **but re-introduces hexavalent chromium**
Trivalent chromium (Cr III / TCP-type)	`warm`	No	Good corrosion, **no Cr VI** — increasingly the preferred "chrome-free" seal
Non-chromate (hot DI water, or proprietary)	`hot DI ~96–100 °C, or per product`	No	Metal-free options available; performance varies by product; verify it meets spec
Unsealed (lightly sealed)	`n/a`	No	**Best paint/primer/bond adhesion**; least bare-corrosion protection — **must be primed/painted promptly**

⚠ **HEADS-UP — THE IRONY OF THE CHROMATE SEAL**

BSAA's whole purpose is to **get hex chrome off the line.** If your shop uses a **dichromate (Cr VI) seal**, you've removed Cr VI from the *anodize* tank but **kept it in the *seal* tank.** Shops pursuing *full* hex-chrome elimination use **trivalent or non-chromate seals**, or leave parts unsealed for paint. Know exactly which seal your line runs and treat a chromate seal tank with full Cr VI controls.

⚙ **TECHNICAL STUFF — WHY SEALING FIGHTS PAINT ADHESION**

A sealed, hydrated, closed-pore surface gives primer **less to grip** (mechanically and chemically) than an open, fresh anodic oxide. That's why aerospace paint/bond specs so often call for **unsealed or only lightly sealed** BSAA — the open pore *is* the adhesion. The trade is real and it is the central judgment of this station.

TECHNICAL STUFF — WHY HOT WATER SEALS AT ALL

Near-boiling DI water hydrates the pore-wall oxide to **boehmite (AlOOH)**, which swells and pinches the pore mouths shut. Chromate and trivalent seals add inhibitor chemistry into the pores for extra corrosion protection. All "sealing" routes do the same basic thing — **close/plug the pores** — by different chemistry, with different corrosion and adhesion outcomes.

⚠ HEADS-UP — HOT SEAL = SCALD HAZARD

A chromate or hot-water seal runs **~90–100 °C.** Splashes and steam cause **serious scald burns.** Face shield, heat-resistant gloves, long sleeves; mind the steam cloud; lower parts gently.

⚠ HEADS-UP — CHROMATE / FLUORIDE SEALS ARE HEALTH & WASTE CONCERNS

A **dichromate seal is Cr VI** — full Cr VI controls and waste treatment (reduce-then-precipitate) for *that* tank. Any **fluoride**-bearing seal is especially hazardous (deep, delayed burns; **calcium gluconate** first-aid). Route chromate/fluoride/nickel seal waste to **proper treatment — never to drain.**

MAKE-UP & KEY PARAMETERS

PARAMETER	RANGE / PRACTICE	NOTES
Seal type	Per spec (chromate / trivalent / non-chromate / none)	The job's callout decides
Water	`DI (hot-water/non-chromate seal)`	Hard water → bloom
Temperature	`Hot seals ~90–100 °C; trivalent warm; unsealed n/a`	Per chosen seal
pH	`Per seal product`	Wrong pH = poor seal/bloom
Time	`Per seal type & coating thickness`	Thin coat seals quickly
Result	Sealed (passes seal test) **or** unsealed (ready to prime)	Match to the paint/corrosion requirement

STEP-BY-STEP

1. **Confirm the spec's seal state** (sealed — and which seal — *or* unsealed for paint). 2. If sealing: confirm seal type, temperature, pH, and DI water are correct. 3. Confirm the part came from a clean cool rinse. 4. Immerse gently (mind the scald), full time. 5. Withdraw, rinse if specified, inspect for **bloom**/contamination. 6. If **unsealed for paint:** keep the surface clean and route to priming **promptly** (per the spec's max time). 7. Move to final rinse/dry.

TEACH-BACK — CAN YOU ANSWER?

- What sealing does (closes pores) and the trade-off (corrosion vs. paint adhesion).
- The four seal options and which contain Cr VI.
- Why a chromate seal "puts the chrome back."
- The scald and chromate/fluoride hazards.
- Why unsealed parts must be primed promptly.

TOP MISTAKES

- Sealing a part that was supposed to stay unsealed for paint (kills adhesion).
- Using a Cr VI seal when the customer wanted chrome-free.
- Hard/tap water → bloom; plunging into a near-boiling tank.
- Routing chromate/fluoride waste to drain.
- Letting an unsealed part sit and age before priming.

SIGN-OFF — STATION 10

"I confirmed the spec's seal state, executed the correct seal (or correctly left the part unsealed for prompt priming), handled the scald and any chromate/fluoride hazards, and understand the corrosion-vs-adhesion trade-off."

Trainee: ____________________ Trainer: ____________________ Seal state/type: ____________

STATION 11 OF 12

RINSE / DRY

"FINISH CLEAN, DRY GENTLY, DON'T UNDO YOUR WORK — AND DON'T CONTAMINATE A PAINT SURFACE."

PURPOSE & THE "WHY"

Give the part a final clean rinse and a **gentle dry** without water spotting, contamination, or thermal damage. A clean final rinse removes any seal-tank drag-out so the part dries spot-free. **If the part is unsealed for paint, cleanliness here is critical** — fingerprints, oils, or hard-water residue on a primer surface cause adhesion failures downstream.

PARAMETER	RANGE	NOTES
Final rinse water	`DI`	Spot-free, contamination-free
Rinse temperature	`Warm (or per spec)`	Aids drying; don't over-heat an unsealed surface
Dry method	`Clean warm forced air; gentle oven`	Avoid excessive heat
Handling	`By rack / clean gloves`	**No bare-hand contact** on paint/bond surfaces

⚠ HEADS-UP

Parts coming off a hot seal are **hot** — burn risk. Let them cool or use gloves. Dry ovens are a heat hazard; mind hot racks.

💡 TIP — PROTECT THE PAINT SURFACE

On an **unsealed** BSAA part headed for primer, **bare-hand contact is a defect** — skin oils contaminate the adhesion surface. Handle by the rack or with clean gloves, keep it clean, and get it to paint promptly.

STEP-BY-STEP

- Final DI rinse.
- Drain.
- Gentle dry.
- Handle by rack/clean gloves to inspection/paint.

TOP MISTAKES

- Tap-water final rinse (spots/contamination).
- Over-aggressive drying.
- **Bare-hand handling of a paint surface.**
- Letting an unsealed part sit/age.

SIGN-OFF — STATION 11

Teach-Back: Why a clean final rinse + gentle dry protects the job; why bare-hand contact ruins a paint surface.

"I gave a clean DI final rinse, dried gently without spotting/overheating, and handled the finished/paint surface cleanly."

Trainee: ____________________ Trainer: ____________________ Date: ______________

STATION 12 OF 12

FINAL INSPECTION

“PROVE IT’S RIGHT — THICKNESS, SEAL/ADHESION STATE, AND A CLEAN PAINT-READY SURFACE.”

PURPOSE & THE “WHY”

Verify the finished part meets spec: **coating thickness, seal state (or seal-quality if sealed), appearance, and — for paint-base parts — a clean, uncontaminated surface ready to prime.** BSAA quality isn’t visible by eye alone. **Thickness** (eddy-current, where applicable to such thin coatings) confirms you grew enough oxide; **seal-quality tests** (if sealed) confirm pore closure; and for unsealed paint-base parts, the key checks are **cleanliness and prompt routing to paint.**

CHECK	METHOD	PASS CRITERIA
Coating thickness	Eddy-current (ASTM B244) where applicable; micro-section if needed	Within spec (thin — often a few µm)
Coating presence/quality	Visual + per-spec checks; process-control coupons	Uniform, continuous, no burns/pits
Seal quality (if sealed)	Dye-stain (ASTM B136) / admittance (ASTM B457) per spec	Per spec for the chosen seal
Paint readiness (if unsealed)	Cleanliness, no contamination, within time-to-paint	Clean, prime promptly
Corrosion (per spec)	Salt spray (ASTM B117) on coupons per spec	Per customer requirement
Appearance	Visual under good light	No burns, pits, streaks, bad contact marks, handling damage
Dimensions	Gauging where critical	Within tolerance after (small) growth

★ **REMEMBER — THE TWO BSAA QUESTIONS**

“Did we grow the right (thin) oxide?” and **“Is the seal/adhesion state correct for how this part is used?”** A BSAA part can look fine and still fail if it was *sealed when it should have been left open for paint* (or vice versa). Verify the **seal state against the spec**, not just the appearance.

💡 **TIP**

Because BSAA coatings are so thin, **thickness measurement and seal/process verification often rely on process-control coupons and salt-spray testing** run alongside the parts, per the customer spec. Know which checks your shop runs on parts vs. coupons.

STEP-BY-STEP

- Thickness on significant surfaces (per method).
- Coating/appearance check.
- Seal-quality test if sealed (or confirm clean & route to paint if unsealed).
- Per-spec corrosion coupon; gauge critical dimensions.
- Accept/reject/rework (strip & re-anodize).

TOP MISTAKES

- Checking only easy surfaces.
- Ignoring the seal-state requirement.
- Bare-hand handling of paint surfaces.
- Letting unsealed parts age past time-to-paint.
- Judging under bad light.

SIGN-OFF — STATION 12

Teach-Back: Which check answers “enough oxide?” vs. “correct seal/adhesion state?”; why a chromate-vs-non-chromate-vs-unsealed mistake is a real failure.

“I verified thickness/coating per method, confirmed the seal state matched the spec, inspected for defects, kept paint surfaces clean, and confirmed critical dimensions.”

Trainee: ____________ Trainer: ____________ Thickness: ________ µm · Seal state: ________

PART THREE — ACROSS THE LINE

THE CROSS-CUTTING TOPICS

ACROSS THE LINE

THE IDEAS THAT DON'T LIVE IN ONE TANK — THEY TIE THE WHOLE LINE TOGETHER

• COATING THICKNESS & DIMENSIONAL GROWTH (DEEP DIVE)

BSAA is a **thin coating by design (~2–7 µm)** — that thinness is a feature for fatigue-critical, tight-tolerance aerospace parts. Unlike Type II (set by current density × time), **BSAA is voltage-controlled**: you ramp to and hold the spec voltage (~15 V) for the spec cycle (~20–25 min); thickness lands in the spec window. **Dimensional growth** still follows the **~50% in / ~50% out** rule, but the numbers are tiny — which is exactly why aerospace likes it. Verify thickness with an **eddy-current gauge** (ASTM B244) on the significant surface where the method applies, and rely on **process-control coupons** for very thin coatings per spec.

> ★ **REMEMBER**
> **Thin on purpose, voltage-controlled, fatigue-friendly.** Don't try to "push it thicker" — that's not what BSAA is for, and over-driving changes the properties aerospace is counting on.

• FATIGUE STRENGTH (DEEP DIVE)

Anodizing removes some base metal (at etch) and grows a hard, relatively brittle oxide — both of which can **reduce fatigue life** of a stressed part. The reason chromic was the aerospace standard is that its **thin coating and gentle chemistry** were kind to fatigue. **BSAA was specifically designed to match that fatigue-friendliness** without Cr VI: **light etch (minimal metal removal), mild buffered electrolyte, thin coating.** That's why **etch discipline (Station 03)** and staying within the spec'd thin coating window matter so much — they protect fatigue life on flight hardware.

> ★ **REMEMBER**
> On aerospace parts, **less is more.** Minimal etch + thin coating = preserved fatigue strength. Over-etching or over-anodizing isn't "more protection" — it's a fatigue debit.

• PAINT / PRIMER & ADHESIVE ADHESION (DEEP DIVE)

This is BSAA's headline job. The **open, porous, fresh anodic oxide** is one of the best **mechanical and chemical keys** for aerospace primers, topcoats, and structural adhesives. Two operator levers control adhesion: **(1) the seal decision** — unsealed or lightly sealed gives the best grip; a full seal trades adhesion for bare-corrosion protection — and **(2) cleanliness and promptness** — keep the surface free of skin oils and contamination and prime within the spec's time window. A perfect anodize that gets fingerprinted or aged before paint can still fail.

> ★ **REMEMBER**
> Two ideas rule this page: **thin coating by design** (don't over-drive) and **adhesion = open pores + cleanliness + promptness** (don't over-seal or contaminate a paint surface).

SEAL OPTIONS — THE CHROMATE-VS-NON-CHROMATE CHOICE (DEEP DIVE)

The seal is where BSAA's "chrome-free" promise is won or lost:

- **Dilute chromate (sodium dichromate):** best-proven corrosion protection, the traditional aerospace seal — **but it is Cr VI**, so it re-introduces the very hazard BSAA was meant to eliminate.
- **Trivalent chromium (Cr III / TCP-type):** good corrosion protection, **no Cr VI** — the leading "chrome-free" seal where the spec allows.
- **Non-chromate (hot DI water or proprietary):** metal-free/low-hazard options; performance varies — must be **qualified to the spec.**
- **Unsealed / lightly sealed: maximum paint and bond adhesion**; least bare-corrosion protection — used when the part is **primed/painted promptly.**

The choice is driven by **(a) does the part get painted?** and **(b) does the customer permit chromate?** It is always a **spec/engineering decision**, never an operator preference.

> ⚠ **HEADS-UP**
> A **chromate seal tank is a Cr VI source** even though the anodize tank isn't — full Cr VI controls and reduce-then-precipitate waste treatment apply to that tank. Using the wrong seal (chromate when chrome-free was required, or sealed when unsealed-for-paint was required) is a real nonconformance.

ALUMINUM ALLOY EFFECTS (DEEP DIVE)

BSAA runs on the **hard aerospace alloys**: **2xxx (copper, e.g., 2024)** and **7xxx (zinc-copper, e.g., 7075)**, plus **clad (Alclad) sheet.** Copper-rich alloys leave heavy smut and are fussier to anodize — BSAA's mild, buffered, low-voltage chemistry handles them more gently than hot concentrated sulfuric. **Clad sheet** has a thin pure-aluminum skin that anodizes beautifully but **can be etched through** — so **light etch discipline** protects the cladding. Rack like-alloy with like-alloy and process a sample of the actual alloy/lot when in doubt. *(Cross-reference Chapter 7.)*

RACKING & ELECTRICAL CONTACT (DEEP DIVE)

The part is the anode, so **the rack is the wire that grows your coating.** Good racking: **firm current-carrying contact**; **contact placed (per drawing) where the bare mark won't matter**; **orientation for gas escape and drainage**; **no parts touching**; **maintained racks** (titanium preferred). Poor contact is the most common cause of thin coatings, burned spots, and dropped parts. *(Cross-reference Station 01.)*

RINSING & WATER/DI MANAGEMENT

Rinses are firewalls between opposite chemistries (caustic etch vs. acid desmut/anodize) and protectors of the seal/paint surface. **Counter-current cascade** rinsing saves water; **conductivity** is the cleanliness gauge (lower = cleaner). Use **DI water** for the post-anodize rinse, seal, and final rinse — tap-water impurities (especially **chloride** and hardness) cause pitting, poor seal, and bloom, and contaminate paint surfaces.

THE FRIENDLY PART

WASTE & ENVIRONMENTAL

WHY BSAA'S WASTE STORY IS SO MUCH BETTER THAN CHROMIC'S

The headline: because the BSAA anodize bath has **no hexavalent chromium**, you avoid chromic anodizing's hardest waste burden — the **reduce-then-precipitate Cr VI treatment** and the carcinogenic-mist air permit. That's a major environmental and compliance win. But a BSAA line still generates regulated streams — **never to a floor drain untreated:**

- **Spent boric-sulfuric anodize & acid rinses** — acidic; neutralize/treat per permit; manage **dissolved aluminum** (precipitated as aluminum hydroxide sludge) and **boron** (boron can be regulated in discharge — check your permit).
- **Spent caustic etch** — high pH; contains dissolved aluminum; neutralize/treat.
- **Desmut waste** — acidic; **fluoride-bearing** desmuts need fluoride treatment (precipitate as calcium fluoride).
- **Seal waste** — depends entirely on the seal: a **chromate (Cr VI) seal** needs **reduce-then-precipitate Cr VI treatment** (the one place Cr VI can persist on a BSAA line); **trivalent/nickel** seals need their own metal treatment; **non-chromate/hot-water** seals are the easiest.
- **Air** — acid mist and steam need exhaust per permit — but **no chromic-acid mist scrubbing** for the anodize tank.

⚠ **HEADS-UP**

Acidic and caustic streams must be **neutralized under control** (never casually mixed). **Boron** may have discharge limits; **fluoride** and any **chromate seal** streams need **dedicated treatment.** Know your shop's segregation scheme — the wrong drum or drain can be a reportable violation, *especially* if you run a chromate seal.

★ **REMEMBER**

BSAA's environmental selling point is real: **no Cr VI in the anodize bath = dramatically simpler waste and air than chromic.** The one asterisk is the **seal** — if it's chromate, the Cr VI burden comes back in that tank.

☀ **FUN FACT**

Switching a chromic line to BSAA can shrink a shop's hazardous-waste and air-permit footprint enough to change its regulatory standing — one reason the "green replacement" was as much an environmental decision as a worker-safety one.

HOW WE PROVE IT

QUALITY CHECKS & TEST METHODS

THE CHECKS THAT CONFIRM A BSAA PART IS RIGHT — THICKNESS, SEAL STATE, AND (ABOVE ALL) ADHESION

CHECK	METHOD (TYPICAL STD)	ANSWERS
Coating thickness	Eddy-current gauge (ASTM B244) where applicable; micro-section	“Enough (thin) oxide grown?”
Coating weight/quality	Per spec; process-control coupons	“Coating within spec?”
Seal quality (if sealed)	Dye-stain (ASTM B136); admittance (ASTM B457) per spec	“Pores closed correctly?”
Paint/bond adhesion	Tape/adhesion tests per customer spec	“Will the primer/adhesive hold?”
Corrosion	Salt spray (ASTM B117) on coupons per spec	“Will it survive service?”
Appearance/defects	Visual under controlled light	Burns, pits, streaks, marks, contamination

💡

TIP

Because BSAA is thin and is usually a **paint base**, the most important real-world checks are often **adhesion** and **salt-spray on coupons**, plus **confirming the correct seal state** — not just a thickness number. Run the checks your customer spec actually requires.

★

REMEMBER

The two BSAA questions in test form: **did we grow the right thin oxide** (thickness/coupon) and **is the seal/adhesion state correct for how this part is used** (seal test if sealed; cleanliness + adhesion if unsealed)? Appearance alone never proves a BSAA part is good.

⚙

TECHNICAL STUFF — WHY COUPONS MATTER SO MUCH ON BSAA

At a few microns thick, a BSAA coating is at the edge of what some thickness gauges resolve reliably, and destructive checks would scrap a flight part. So shops run **process-control coupons** of the same alloy through the same load, then test *those* for thickness, seal, adhesion, and salt-spray — proving the process without harming the hardware.

FIELD REFERENCE

TROUBLESHOOTING QUICK-INDEX

FIND YOUR SYMPTOM, READ THE LIKELY CAUSE, TRY THE FIX

DEFECT / SYMPTOM	LIKELY CAUSE	FIX
Thin or no coating	Poor contact; **wrong polarity (part not anode)**; low voltage; weak bath	Re-rack tight; **verify part = anode**; check voltage/bath
Powdery / soft coating	Bath too warm / out of balance	Bring temp & chemistry into range; check agitation
Pitting	**Chloride or fluoride contamination**; impurities	Lab-check Cl^-/F^-; switch to DI; treat bath
Burned/rough at contact/edges	Current crowding; poor contact; chloride	Improve racking; check chloride; check ramp
Dull / gray / streaky	Copper alloy (2024/7075); smut not removed; high dissolved Al	Confirm alloy; verify desmut; lab-check Al
Etched-through cladding	Over-etch on clad sheet	Lighten/shorten etch; protect cladding
Poor paint/primer adhesion	Over-sealed for paint; contamination; aged surface; over-etch	Leave unsealed per spec; prime promptly; keep clean
Poor corrosion (bare part)	Unsealed when it needed sealing; wrong seal	Apply spec'd seal; verify seal state
Sealing bloom (chalky film)	Hard water; wrong pH; over-concentrated seal	DI water; correct pH/concentration; light post-dip
Cr VI where you didn't expect it	Chromate seal in use	Confirm seal type; apply Cr VI controls to seal tank
Voltage won't hold / current odd	Poor contact; failing cathodes; bath imbalance	Check contacts/cathodes; lab-check bath

TIP

Most BSAA defects trace to one of five roots: **contact/polarity** (thin/no coat), **contamination** (chloride/fluoride — pitting), **alloy/desmut** (gray/dull), **etch discipline** (clad/fatigue), or **the seal decision** (corrosion vs. adhesion). Name the root, not just the symptom.

★ REMEMBER

Two are **safety-relevant** as well as quality: **wrong polarity** (verify the part is the anode) and **using a chromate seal without Cr VI controls.** When in doubt about a bath or seal change, ask your supervisor before you act.

EVERY TERM, PLAIN AND SIMPLE

GLOSSARY

PLAIN-LANGUAGE DEFINITIONS FOR EVERYTHING IN THIS MANUAL

Admittance test: an electrical test (ASTM B457) of how “open” a coating still is — used to confirm sealing.
Anode: the **positive** electrode. In anodizing it’s **your part** — where the oxide grows.
Anodizing: growing an integral oxide layer out of a metal’s surface by making the part the anode in an electrolyte.
Barrier layer: the thin, dense oxide at the bottom of the coating, against the metal.
Boehmite: hydrated aluminum oxide (AlOOH) formed during hot-water sealing; its swelling closes pores.
Boric acid (H_3BO_3): a weak acid that **buffers/moderates** the BSAA electrolyte; low acute hazard but classified **toxic for reproduction** (REACH/CLP) — control dust/mist/ingestion.
BSAA: Boric-Sulfuric Acid Anodizing — the **no-Cr-VI, boric-sulfuric replacement for chromic (Type I)** anodizing; MIL Type IC; Boeing BAC5632.
Cathode: the **negative** electrode (stainless/lead/aluminum here) — carries current and bubbles off hydrogen; does **not** feed the coating.
Chromate seal: a dilute dichromate (**Cr VI**) seal — excellent corrosion, but re-introduces hexavalent chromium.
Clad / Alclad: aluminum sheet with a thin pure-Al skin over a copper-rich core; anodizes well but can be **etched through.**
Cr VI (hexavalent chromium): the carcinogenic chromium in chromic acid — **absent from the BSAA bath** (but present if a chromate seal is used).
Desmut / deox: an acid dip that removes the dark **smut** left by caustic etching.
Dimensional growth: the coating grows **~half in, half out**; tiny on thin BSAA.
Drag-out / drag-in: solution clinging to a part leaving a tank (out) / contaminating the next tank (in).
Eddy-current gauge: non-destructive instrument for measuring coating thickness on aluminum.
Etch: a hot **caustic** dip; on BSAA it is kept **light** to protect fatigue/tolerance/cladding.
Fatigue strength: a metal’s resistance to cracking under cyclic load — preserved by BSAA’s light etch + thin coating.
Integral coating: a coating that *is* part of the metal (grown from it), not added on top.
MIL-PRF-8625 (MIL-A-8625): the spec defining anodize Types I, **IC**, II, III. **Type IC = non-chromic substitute (BSAA/TSA).**
PAA: Phosphoric-Acid Anodize — chrome-free pretreatment optimized for **adhesive bonding** (next manual).
Porous layer: the upper oxide full of vertical **pores** — on BSAA, the key to **paint/primer adhesion.**
Sealing: closing the pores (chromate / trivalent / non-chromate / hot water) — boosts corrosion but can reduce paint adhesion.
Smut: dark residue of alloying elements (especially **copper** on aerospace alloys) left after caustic etching.
Trivalent chromium (Cr III / TCP): a **chrome-free** (no Cr VI) seal/conversion chemistry.
TSA: Thin-film Sulfuric Anodize — a dilute-sulfuric, chrome-free aerospace pretreatment; close sibling of BSAA.
Type IC: the MIL-PRF-8625 class for **non-chromic** anodizing — **BSAA’s designation.**
Valve metals: metals that anodize to protective oxides (aluminum, titanium, magnesium, tantalum, niobium).
Water-break test: a free cleanliness check — clean metal sheets water; dirty metal beads it.

★ **REMEMBER**

Five terms matter most: **anode** (your part), **boric acid** (the buffer that makes a sulfuric bath act like chromic), **Type IC / no Cr VI** (why BSAA exists), **paint base / open pores** (its real job), and **the seal choice** (corrosion vs. adhesion, chromate vs. chrome-free). Master those and you understand BSAA.

☀ **FUN FACT**

The word **“anode”** comes from the Greek *ánodos*, “a way up” (Faraday’s term for the electrode where current enters the cell). On this line, that “way up” is literally where your coating grows.

TEMPLATE — PRINT ONE PER OPERATOR

OPERATOR TRAINING MATRIX

HOW A SUPERVISOR PROVES — ON PAPER — THAT EACH OPERATOR IS TRAINED AND SIGNED OFF BEFORE RUNNING SOLO

Sign-off levels: O = Observed only · **A** = Assisted under supervision · **Q** = Qualified to run solo · **T** = Qualified to Train others.

#	STATION / SKILL	O/A/Q/T	TRAINER INIT & DATE	OPERATOR INIT & DATE	RE-CERT DUE
1	**Safety / PPE / SDS / hygiene — acid, boric acid, caustic, hot/chromate seal, hydrogen, DC (all stations)**	____	________	________	______
2	Station 01 — Inspection & Racking (contact)	____	________	________	______
3	Station 02 — Clean / Degrease + water-break	____	________	________	______
4	Station 03 — Etch (LIGHT — fatigue/clad discipline)	____	________	________	______
5	Station 04 — Rinse + conductivity	____	________	________	______
6	Station 05 — Desmut / Deox	____	________	________	______
7	Station 06 — Rinse	____	________	________	______
8	Station 07 — Anodize (part = anode; boric-sulfuric; ~15 V)	____	________	________	______
9	Voltage/time control & dimensional growth (thin coat)	____	________	________	______
10	Station 08 — Post-anodize rinse (open pores)	____	________	________	______
11	Station 09 — Dye (normally skipped — spec check)	____	________	________	______
12	Station 10 — Seal decision (chromate/trivalent/non-chromate/unsealed)	____	________	________	______
13	Station 11 — Rinse / Dry (paint-surface cleanliness)	____	________	________	______
14	Station 12 — Final Inspection (thickness/seal state/adhesion)	____	________	________	______
15	LOTO on rectifier / electrical safety	____	________	________	______
16	Emergency response — eyewash, shower, spill, fluoride/chromate first-aid	____	________	________	______
17	Waste segregation & environmental controls (incl. boron & any chromate seal)	____	________	________	______
18	Troubleshooting — symptom recognition	____	________	________	______

Operator name: ______________________________ **Hire/start date:** __________________

Supervisor name: ______________________________ **Review cycle:** ☐ Annual ☐ Other: _____________

⚠ **HEADS-UP**

Safety rows (1, 15, 16, 17) are not optional and not “fill in later.” An operator may not work *any* station until PPE/SDS/hygiene (including **boric-acid** and any **chromate-seal** controls), LOTO, emergency response, and waste-control awareness are signed. Safety competency precedes production competency.

PART FOUR — CERTIFICATION
20 QUESTIONS · PASSING SCORE 80% (16/20)

COMPLETION TEST

COVERS EVERY STATION PLUS SAFETY — ANSWER IN YOUR OWN WORDS WHERE ASKED

★ **REMEMBER**
Passing score: 80% (16 of 20). Any **safety** question answered wrong (**Q3, Q9, Q13, Q17, Q19**) is an automatic fail of the safety gate — re-teach and re-test before sign-off, regardless of total score. The Answer Key follows — no peeking until you're done!

Name: ______________________ **Date:** ______________ **Score:** ________ / 20 **Safety-gate Qs: 3, 9, 13, 17, 19**

Q1. Anodizing differs from electroplating mainly because in anodizing the part is the:
A. Cathode, and a different metal is deposited on it B. Cathode, and it dissolves C. Anode, and an oxide is grown out of the part itself D. Insulator, and nothing happens

Q2. The BSAA anodize electrolyte is:
A. Boric acid plus a small amount of sulfuric acid B. Chromic acid (hexavalent chromium) C. Concentrated sulfuric acid (~165–200 g/L) only D. Sodium hydroxide

Q3. [SAFETY GATE] Which statement about BSAA's safety is *correct*?
A. BSAA is completely hazard-free B. BSAA contains Cr VI just like chromic anodizing C. The biggest hazard on a BSAA line is hydrofluoric acid in the anodize tank D. BSAA removes the Cr VI carcinogen from the anodize bath, but still has sulfuric acid, boric acid, hot caustic etch, hydrogen, and DC hazards to respect

Q4. Typical BSAA coating thickness is about:
A. 25–100 µm (hardcoat range) B. 2–7 µm (a thin coating) C. 0.5–1 mm D. There is no coating

Q5. On a BSAA line, the porous oxide is used mainly to:
A. Hold deep decorative dye color B. Make the part electrically conductive C. Grip primer, paint, and adhesive (a paint/bond base) D. Add significant thickness

Q6. (Short answer) Explain in your own words why anodizing is **not** plating — name which electrode the part is and **where the coating comes from.**

Q7. "Dimensional growth" in anodizing means the coating grows roughly:
A. Half into the surface and half out (each surface moves out ~half the thickness) B. All outward C. All inward D. It shrinks the part

Q8. For a typical BSAA aerospace part that will be **painted**, the usual practice is:
A. Dye it a bright color, then seal heavily B. Apply Type III hardcoat first C. Always apply a thick decorative coating D. Skip dye; often leave it unsealed (or lightly sealed) and prime promptly for maximum adhesion

Q9. (Short answer) [SAFETY GATE] Name the **two corrosive hazards** on the line (one high-pH, one low-pH), and state the universal rule for adding acid during bath make-up.

Q10. Typical BSAA anodize voltage and temperature are about:
A. 200 V; 0 °C B. ~15 V; ~26–30 °C C. 40 V; held cold near 0 °C D. 1 V; 90 °C

Q11. A 2024 or 7075 part comes out **dull/gray** even though the line is running fine. The most likely cause is:
A. The bath is too clean B. Too much dye C. A copper-rich aerospace alloy (copper dulls the coating) and/or smut not fully removed D. The rectifier is off

Q12. (Short answer) What is **smut**, where does it come from, and which station removes it? (Note why it's heavier on aerospace alloys.)

Q13. [SAFETY GATE] Boric acid in the BSAA bath is best described as:
A. Low in acute hazard, but classified as toxic for reproduction (REACH/CLP) — so control dust, mist, and ingestion B. A confirmed carcinogen worse than Cr VI C. Completely non-hazardous and safe to eat D. Explosive on contact with water

Q14. Why was BSAA developed?
A. To make a thicker, harder coating than hardcoat B. To replace chromic-acid anodizing and eliminate hexavalent chromium while keeping thin, paint-ready, fatigue-friendly properties C. To allow brighter decorative dye colors D. To run at much higher voltage than chromic

Q15. BSAA coating thickness is controlled mainly by:
A. Dye color B. Seal temperature C. The alloy only D. The anodizing voltage and time (it is a voltage-controlled process)

Q16. (Short answer) Name the **seal options** for BSAA and the key trade-off the operator must understand. (Include which option re-introduces Cr VI, and why a part might be left **unsealed.**)

Q17. (Short answer) [SAFETY GATE] List **three** pieces of PPE required at the BSAA acid/desmut tanks, and name the special first-aid concern if your desmut or seal contains **fluoride.**

Q18. Under MIL-PRF-8625, BSAA is classified as:
A. Type II B. Type III C. Type IC (non-chromic acid anodizing) D. Type I (chromic)

Q19. (Short answer) [SAFETY GATE] Name the **flammable gas** produced at the cathode during anodizing and **two** controls that keep it from becoming a fire hazard.

Q20. (Short answer) Give **two** reasons aerospace adopted BSAA in place of chromic anodizing.

⚠ **SAFETY SECTION — MUST BE CORRECT TO CERTIFY**

Questions 3, 9, 13, 17, and 19 cover hazards that can permanently harm you or others. A miss on any safety item means re-teach and re-test before sign-off.

End of test. Hand to your trainer for grading.

FOR TRAINERS & SELF-CHECKERS

ANSWER KEY

SHORT-ANSWER RESPONSES DON'T NEED TO MATCH WORD-FOR-WORD — LOOK FOR THE KEY CONCEPTS IN BOLD

⚠ **HEADS-UP**

Keep this page out of the trainee's copy. Questions **3, 9, 13, 17, and 19** are safety-critical — any wrong answer there requires re-training on that topic before sign-off, regardless of total score.

Q1. C — The part is the **anode**, and oxide is grown out of the part itself.

Q2. A — **Boric acid + a small amount of sulfuric acid** (~5–10 g/L boric, ~30–50 g/L sulfuric).

Q3. D — BSAA **removes the Cr VI carcinogen** from the anodize bath, but **sulfuric acid, boric acid, hot caustic etch, hydrogen, and DC hazards remain** — it is safer/greener, not hazard-free.

Q4. B — ~2–7 µm (a thin coating by design).

Q5. C — The pores **grip primer/paint/adhesive** — BSAA is a paint/bond base, not a decorative dye coating.

Q6. In plating the part is the **cathode** and a **different metal is deposited onto it** from the bath. In anodizing the part is the **anode**, and the coating is **aluminum oxide grown out of the part's own surface** (oxygen from the bath combines with the part's aluminum) — nothing is deposited from elsewhere.

Q7. A — Half in, half out; each surface grows out ~half the thickness (tiny on thin BSAA).

Q8. D — Skip dye; **often left unsealed (or lightly sealed) and primed promptly** for maximum paint/bond adhesion.

Q9. The **caustic etch** (high pH / sodium hydroxide) and the **sulfuric desmut/anodize** (low pH / acid) — both cause severe burns. Universal rule: **always add ACID to WATER, never water to acid.**

Q10. B — ~15 V; ~26–30 °C (lower voltage and warmer than a cold Type II tank).

Q11. C — A **copper-rich alloy (2024/7075)** dulls the coating, and/or **smut wasn't fully removed** at desmut.

Q12. Smut is the dark film of alloying elements — especially **copper** on aerospace alloys, plus silicon/iron — **left behind by the caustic etch** when aluminum dissolves. It's removed at the **desmut/deox** station (acid dip). It's heavier on **copper-rich 2xxx/7xxx** alloys.

Q13. A — Boric acid is **low in acute hazard but classified toxic for reproduction (REACH/CLP)** — control **dust, mist, and ingestion** (hygiene, no eating/drinking). Not a carcinogen, not in Cr VI's league, but **not harmless.**

Q14. B — To **replace chromic-acid anodizing and eliminate hexavalent chromium** while keeping the thin, paint-ready, fatigue-friendly aerospace properties.

Q15. D — The **anodizing voltage and time** — BSAA is a **voltage-controlled** process (ramp to ~15 V and hold for the cycle).

Q16. Options: **dilute chromate** (best corrosion but **re-introduces Cr VI**), **trivalent chromium (Cr III/TCP)** (good corrosion, **no Cr VI**), **non-chromate / hot DI water** (low-hazard, performance varies), and **unsealed/lightly sealed** (best **paint/bond adhesion**, least bare-corrosion). Key trade-off: **sealing boosts corrosion of a bare part but reduces paint adhesion** — so paint-base parts are often left unsealed and primed promptly. The choice is set by **the spec**, not the operator.

Q17. PPE (any three): sealed chemical splash goggles, face shield, chemical-resistant gloves, apron/suit, chemical-resistant boots. **Fluoride concern:** fluoride causes **deep, delayed, systemically toxic burns** — needs specific first-aid (e.g., **calcium gluconate** per SDS/medical plan) and immediate medical attention; ordinary flushing alone may not be enough.

Q18. C — **Type IC** (non-chromic acid anodizing, the substitute for Type I/IB) under MIL-PRF-8625.

Q19. Hydrogen gas (at the cathode). Controls (any two): **ventilation/exhaust running**, **no open flames/sparks/grinding nearby**, no smoking, keep ignition sources away.

Q20. Any two of: **eliminates the Cr VI carcinogen**; **keeps thin, paint-ready coating**; **minimal effect on fatigue strength**; **good corrosion protection**; **minimal dimensional change**; **lower energy**; **drop-in for many chromic callouts**; **simpler waste/air** than chromic.

Answer-key letter distribution (MC): A: Q2, Q7, Q13 · B: Q4, Q10, Q14 · C: Q1, Q5, Q11, Q18 · D: Q3, Q8, Q15 — a healthy spread across all four letters.

★ **REMEMBER**

A score of 16/20 passes — but every safety question (3, 9, 13, 17, 19) must be correct to certify. Miss a safety item → re-teach and re-test before sign-off. We don't gamble with people.

PLATING POSTERS

PLATING POSTERS INC · METAL FINISHING REFERENCE SERIES

CERTIFICATE OF COMPLETION

BORIC-SULFURIC ACID ANODIZING (BSAA) TRAINING PROGRAM

This certifies that

____

OPERATOR NAME

has successfully completed the **Boric-Sulfuric Acid Anodizing (BSAA)** training course — covering the full anodize workflow (Inspection & Racking, Clean, Light Etch, Rinse, Desmut/Deox, Anodize, Dye [normally skipped], Seal, Rinse/Dry, and Final Inspection), the fundamental concept that **the part is the anode and the oxide is grown out of the part itself**, the identity of BSAA as the **boric-sulfuric, no-hexavalent-chromium replacement for chromic anodizing (MIL Type IC / Boeing BAC5632)**, the thin paint-base oxide and dimensional growth, voltage-and-time thickness control, electrical-contact and racking practice, aerospace aluminum-alloy and clad-sheet effects, fatigue and paint-adhesion considerations, the **seal-state decision (chromate vs. trivalent vs. non-chromate vs. unsealed for paint)** and seal-quality testing, and — above all — the **safety practices** required of every operator (sulfuric-acid and caustic-etch burn control, boric-acid hygiene and reproductive-toxicity awareness, hot/chromate-seal and fluoride controls, hydrogen and DC-electrical controls, spill response, and waste segregation) — and has passed the Completion Test with a score of ________ / 20, including all required safety competencies.

This operator is hereby cleared for **independent work** at the certified stations listed on the Operator Training Matrix.

Date of completion: ________________ **Re-certification due:** ________________

____

TRAINEE SIGNATURE

____

CERTIFYING TRAINER

____

SUPERVISOR

Training reference only. Verify all parameters against your chemistry supplier's TDS, customer/aerospace specifications (e.g., MIL-PRF-8625 Type IC, Boeing BAC5632, applicable AMS/customer specs), and applicable regulatory requirements before production use. BSAA contains no hexavalent chromium in the anodize bath, but sulfuric acid and caustic etch cause severe burns, boric acid carries a reproductive-toxicity classification, and some seals run hot or contain chromate/fluoride — always consult current SDS and comply with OSHA, EPA, REACH, and local regulations.